Variants of the coil support structure of a stellarator

Catalogue and comparison of all computed designs – strength, package, superconductor, field accuracy and 3D models in PicoGK

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Hobby project, born of curiosity. No connection to any research institute or company, not peer reviewed. All numbers are estimates from simplified models (linear elastic, rigid bearings, magnetic force only) and may contain errors. Calculations, program code and draft text were produced with the support of AI agents; the human set goal and direction and decided at the forks. All figures are own calculations and renderings.

What this is about. This report presents *all* variants of the coil support structure computed for the compact 1-GW stellarator (radius $R_0 = 10.4\,\mathrm{m}$, field $B_0 = 6.5\,\mathrm{T}$, REBCO coils) and compares them: five support tests with identical coils, 199 designs of a design study, the recomputations with finer mesh, a first series "weakening selectively" and the proposals of a surrogate model. For the best designs and for all geometrically representable ones there are 3D models built entirely in PicoGK. It complements the chapter "Strength" of the project report, which shows only the best designs and those close to the limit.

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1 Summary

1. **The support structure decides.** With identical coils, a better structure alone (bearings inside and outside, many stiff connectors, thick walls) lowers the steel stress from 2 129 MPa to 481 MPa (section 3).
2. **7 of 199 designs meet all limits** – after a clean-up: eleven designs had been computed with a coarser mesh for memory reasons, and their connectors unnoticed became one third longer. With the standard mesh two of the originally nine admissible designs drop out (section 4).
3. **16 coils are not enough**; all admissible designs have 20 or 24 coils. More coils buy field accuracy, which can be traded for smoother, shorter coils.
4. **The stress depends mainly on the coil shape**, not on connector count or stiffness: the most important variable is the coil–plasma distance (further away means more strongly deformed coils), then the coil length (section 5). A surrogate model predicts package, nuclear heating, mass and field error well, the stress only poorly.
5. **The stress peaks sit at the ends of the connectors.** A finer mesh along the coil raises the 99-% value by 34–58 % for four of five candidates; a finer resolution of the wall thickness changes almost nothing. Design 0169 is robust: 308 MPa on the standard mesh, 319 MPa on the fine one (section 6).
6. **Material:** 1.4429 / X2CrNiMoN17-13-3 (AISI 316LN). At room temperature the handbook gives a yield strength of only 295 MPa, at 4 K 1 070 MPa was measured. The limit used, 500 MPa, is two thirds of the guaranteed 4-K yield strength of ITER grade C3 (section 7).
7. **Softer connectors make it worse**: making all connectors of 0169 softer raises the case stress (up to +16 %); the user's idea must be applied locally (section 8).
8. **The surrogate finds admissible designs**: three of its four proposals meet all limits; the best is more robust than 0169 on the standard mesh but loses on the fine mesh; 0169 remains the most robust (section 9).
9. **3D models in PicoGK** agree for winding pack and case with the computational model to 0.5–1 % and confirm independently: no case overlaps another case or the package. For 0169 they revealed that neighbouring connectors partly interpenetrate (section 10).

2 How the variants were created

Every design passes through the same chain (program `stellarator/kopplung.py`):

1. **Coils** (simsopt): for a given number of coils, minimum distance to the plasma, allowed length and force target, coil paths are sought that create the magnetic field of the prescribed plasma shape. The figure of merit is the field error $B \cdot n$: the fraction of the field that passes through the plasma boundary instead of running along it.
2. **Structure** (Elmer): around each coil a steel case with winding pack (superconductor) and potting, with connectors in between; bearings inside (towards the plant centre) and outside. The load is the Lorentz force of the current in its own and the other coils' field.
3. **Package**: how much space remains between plasma boundary and case for first wall, blanket, shield and vacuum vessel? This gives nuclear heating and neutron fluence in the winding pack.
4. **Superconductor**: field at the conductor, utilisation and temperature margin of the REBCO tape.
5. **Assessment** against eight limits (table 1) and check whether cases penetrate each other.

Limit	value	origin
steel σ99	$\leq$ 500 MPa	2/3 of the 4 K yield strength (316LN, ITER grade C3)
winding-pack strain	$\leq$ 0.3 %	REBCO tape, Barth 2015: irreversible above 0.45 %
package space	$\geq$ 1 m	first wall + blanket + shield + vessel (ARIES-CS order)
nuclear heating	$\leq$ 2000 W/m^3	ARIES-CS 2 mW/cm^3; EU-DEMO strict 50 W/m^3
neutron fluence	$\leq 3 \cdot 10^{18}$ n/cm^2	REBCO damage (ARC, Sorbom 2015), 20 full-power years
SC utilisation	$\leq$ 0.7 –	operating / critical current at 20 K
temperature margin	$\geq$ 5 K	distance to current-sharing temperature
field error B·n	$\leq$ 3 ‰	mean field error on the plasma boundary

Table 1: Limits against which every design is checked.

Terms.

99-% stress σ_{99} (called "p99" in earlier versions) the von Mises equivalent stress not exceeded in 99 % of the steel volume. The remaining per cent are mostly local peaks at edges. The value is intuitive but not a code quantity; therefore the stress categories are added here.

Stress categories Pressure-vessel codes (ASME) and the ITER magnet criteria classify stresses by their effect. Lines are laid across the wall and the stress along them is decomposed: the mean is the *membrane stress* P_m (the wall is pulled or pushed evenly), the linear part the *bending stress* P_b, the rest the *peak*. Allowed are $P_m \leq S_m$ and $P_m + P_b \leq 1.5\, S_m$; peaks are limited by fatigue and fracture mechanics. Local membrane stresses at discontinuities (joints, bearings) may go up to $1.5\, S_m$.

Reserve distance to the limit in per cent of the limit. 30 % reserve at 500 MPa means 350 MPa.

Robustness the smallest reserve of a design over all limits (user's goal: most robust, not lightest design – "the weight is not so important, it is a building, after all").

Surrogate model a fast approximating function (here a Gaussian process) that learns from the computed designs how a quantity depends on the design variables, and states how uncertain it is.

Voxel a small cube; PicoGK describes bodies as a set of such cubes and can thereby unite, subtract and offset bodies very robustly.

3 Support tests A to E

Before the design study, only the support structure was changed with identical 16 coils. The force analysis had shown that the coils are bent as a whole by the centring force (net force towards the plant centre) when supported outside.

	variant	σ_{99} MPa	max. MPa	deformation mm	WP strain %	steel kt
A	bearing outside (previous), 6 connectors per pair	2 129	8 454	399	0.76	15.6
B	bearing inside	1 454	8 769	273	0.55	15.6
C	bearing inside and outside	1 372	8 692	226	0.54	15.6
D	+ 12 connectors per pair, steel (200 GPa)	947	4 033	107	0.41	23.1
E	+ thickest walls of the range	481	2 455	41	0.31	37.0

Table 2: Support tests with the 16 coils of the starting design. WP = winding pack.

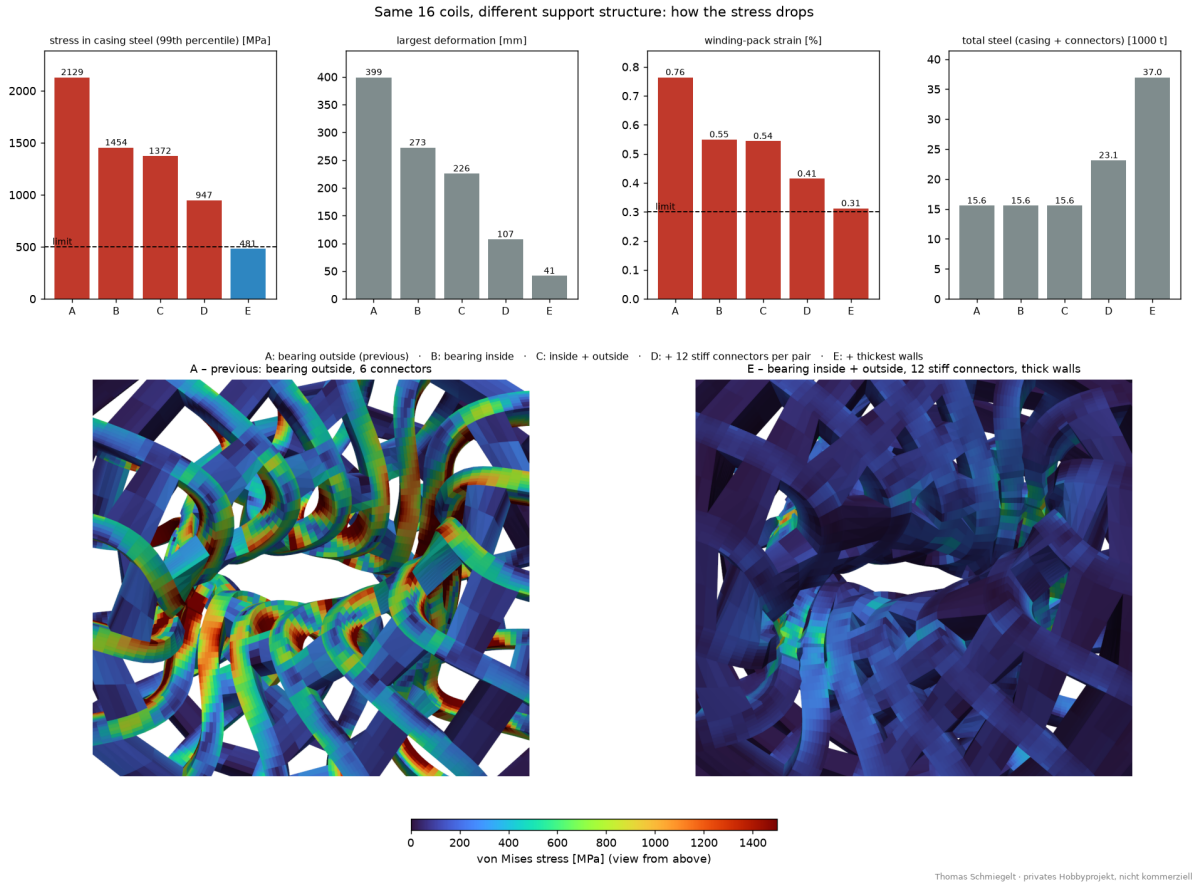


Figure 1: Support tests A to E.

How to read the figure: Top: the key figures of the five tests as bars; red is above the limit. Bottom left test A from above (red: above 1 200 MPa, mainly at the inner legs), bottom right test E: almost everything blue. Each step acts on a different cause: the inner bearing takes the centring force on a short path, connectors shorten the free bending length, thick walls raise the section modulus.

4 Design study doe1

In 200 designs (Latin hypercube) eleven design variables were changed simultaneously: number of coils (16, 20, 24), coil–plasma distance, length per coil, force weight, four wall thicknesses, size of the bearing areas, number (6–14 per pair) and stiffness (20–200 GPa) of the connectors. One design failed in meshing, 199 were computed completely, each in 1–4 minutes.

Clean-up. Eleven designs did not fit into memory with the standard mesh and were computed with 25 % fewer stations per coil. The mesh study revealed that this makes the connectors longer: a connector always spans four stations, and coarser stations are longer. These eleven designs were therefore different designs with larger connectors – and too favourable. All eleven were recomputed with the standard mesh and more memory (marked * in the tables). 0063 (369 →595 MPa) and 0136 (479 →577 MPa) dropped out of the admissible list; 0169 stayed admissible at 308 MPa.

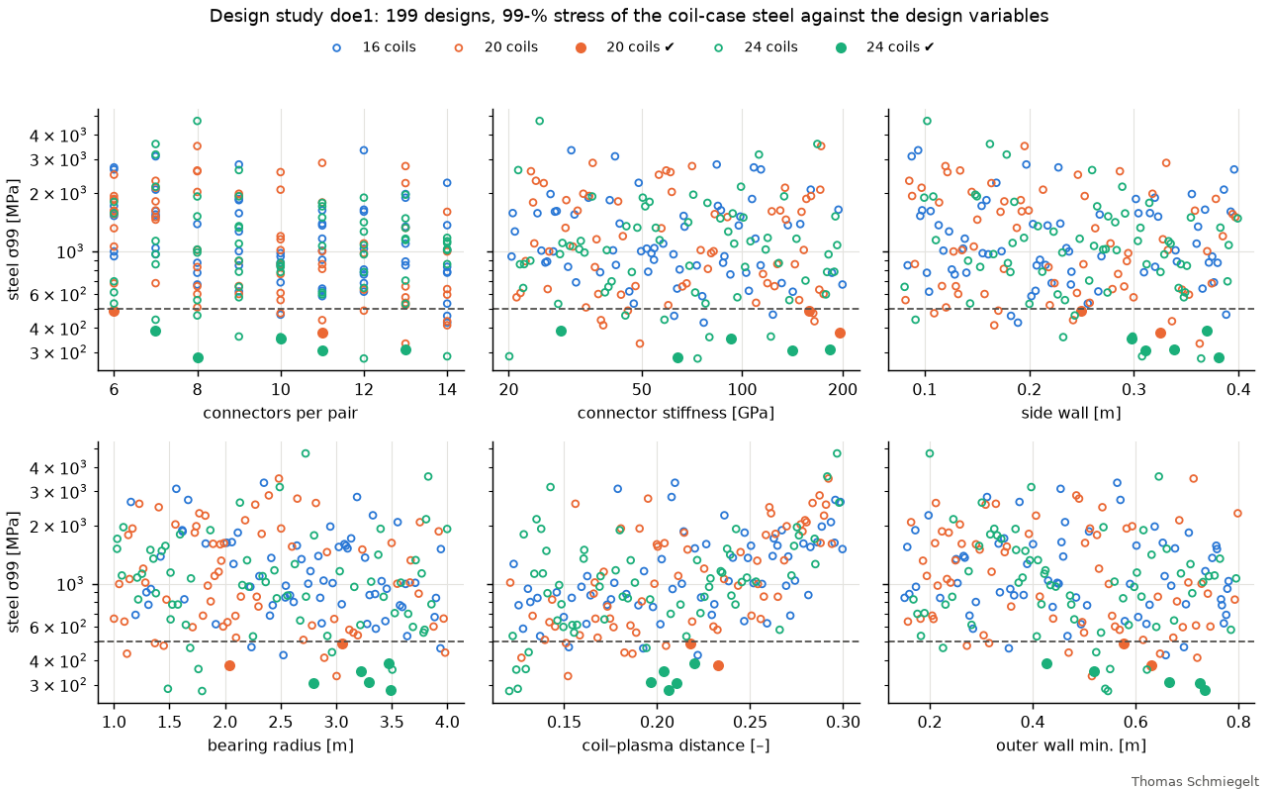


Figure 2: All designs: 99-% stress in the case steel against six design variables.

How to read the figure: Each circle is a computed design, the colour shows the number of coils (blue 16, orange 20, green 24), filled circles meet all eight limits. The dashed line is the limit of 500 MPa, the axis is logarithmic. No panel shows a clear straight line: the stress does not depend on any single variable. The clearest trend is for the coil–plasma distance (bottom centre): further away, higher stress.

What fails most often. The steel stress is the hard bottleneck: only about 11 % of the designs stay below 500 MPa. The winding-pack strain is met by 24 %, the field error by 64 %, package and nuclear heating by more than 90 %, utilisation and temperature margin of the superconductor by all. With 16 coils only 11 of 65 designs reach the field error and 3 the stress limit, none both.

No.	coils	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clearance m	steel kt	smallest reserve (bottleneck)
admissible								
0061	24	351	725	0.17	1.49	1.62	19	30 % (steel σ99)
0169*	24	308	717	0.15	2.13	1.39	28	29 % (package space)
0093	24	283	1 190	0.16	1.45	1.37	28	27 % (package space)
0008	24	305	1 695	0.17	2.20	1.48	26	27 % (field error B·n)
0014	20	377	1 419	0.21	2.21	1.65	28	25 % (steel σ99)
0017	24	389	1 721	0.20	1.66	1.30	23	20 % (package space)
0157	20	492	1 658	0.22	2.37	1.35	18	2 % (steel σ99)
narrowly missed								
0173	24	533	1 827	0.24	1.85	1.28	27	−7 % (steel σ99)
0103	20	440	738	0.19	2.66	1.25	21	−8 % (neutron fluence)
0044	20	540	1 583	0.23	2.76	1.35	26	−8 % (steel σ99)
0004	20	558	1 085	0.26	1.81	1.54	19	−12 % (steel σ99)
0136*	20	577	2 238	0.30	1.62	1.84	26	−15 % (steel σ99)
0022	20	600	2 183	0.26	0.70	1.57	24	−20 % (steel σ99)
0137	24	612	2 562	0.29	0.94	1.57	16	−22 % (steel σ99)
0063*	20	595	4 265	0.37	0.75	1.53	41	−24 % (winding-pack strain)
0042	20	632	3 962	0.34	2.66	1.61	21	−26 % (steel σ99)
0043	24	653	1 208	0.29	1.79	1.30	16	−31 % (steel σ99)
0140	20	656	4 169	0.34	1.46	1.76	22	−31 % (steel σ99)
0085	20	608	2 067	0.28	2.00	1.23	30	−31 % (neutron fluence)

Table 3: Ranking: admissible designs (cleaned) by smallest reserve and the twelve that fail most narrowly. * = recomputed with the standard mesh.

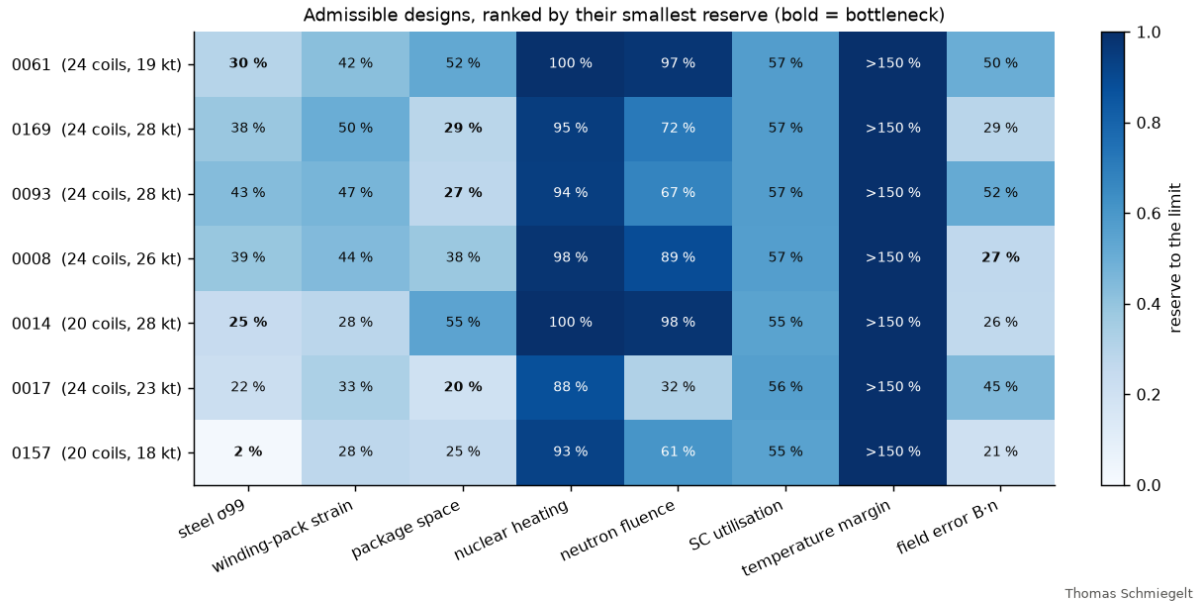


Figure 3: Reserves of the admissible designs per limit.

How to read the figure: Each row is an admissible design, each column a limit. The number is the distance to the limit in per cent; the darker, the more reserve. Bold is the bottleneck, i.e. the smallest reserve, by which the rows are sorted. Stress, package and field error alternate as the bottleneck; superconductor and nuclear heating have plenty of reserve everywhere.

5 What determines the stress: surrogate models

For six quantities a Gaussian process [10] was trained and checked by five-fold cross-validation: each design is predicted by a model that has not seen it.

Quantity	R^2 (CV)	95-% band	most important variables (share)
steel σ99	0.46	91 %	coil–plasma distance (61 %), length per coil (26 %), bearing radius (4 %)
winding-pack strain	0.43	89 %	coil–plasma distance (77 %), length per coil (17 %), bearing radius (2 %)
field error B·n	0.90	88 %	coils per half period (49 %), length per coil (35 %), force weight (11 %)
package space	0.98	92 %	coil–plasma distance (79 %), inner wall (16 %), coils per half period (3 %)
nuclear heating	0.98	92 %	coil–plasma distance (79 %), inner wall (16 %), coils per half period (3 %)
steel mass	0.95	94 %	connectors per pair (37 %), outer wall min. (29 %), inner wall (13 %)

Table 4: Quality of the surrogate models and most important design variables. Share = main effect of the variable on the variation of the prediction.

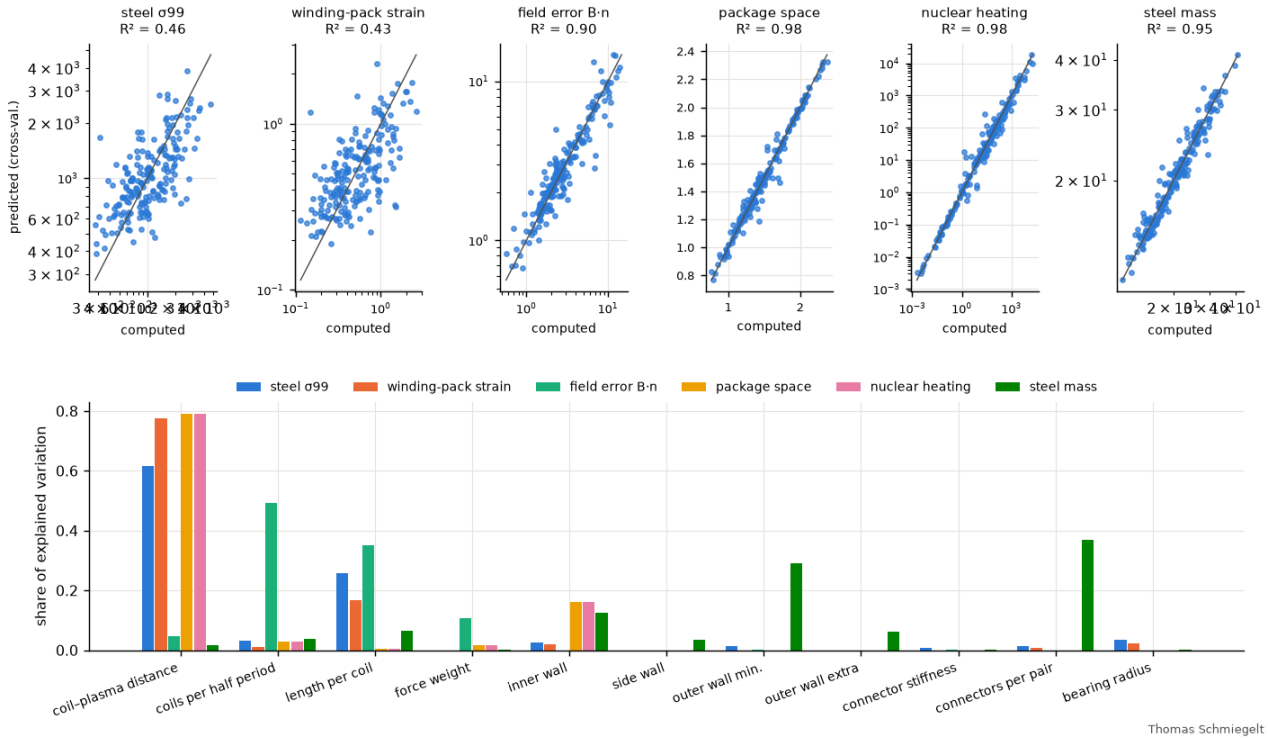


Figure 4: Surrogate models: prediction against computation (top) and influence of the design variables (bottom).

How to read the figure: Top: each point is a design; horizontally the computed value, vertically the prediction of a model that did not know this design. If the points lie on the diagonal, the model is good (R^2 close to 1). Package, nuclear heating, mass and field error lie close to the diagonal, stress and strain scatter widely. Bottom: how strongly each design variable moves each quantity. For the stress (blue) the coil–plasma distance stands out.

Interpretation. It was expected (event log F-123, F-136) that connector number and stiffness determine the stress and that a surrogate predicts it well. Both are refuted. The stress depends on the shape of the coils: coils further from the plasma must be more strongly twisted to create the same field shape, and need more current. On top there is noise from the placement of the connectors: exactly where a connector sits decides the local peak. The rank correlation (Spearman) confirms this without any model: distance +0.50, length +0.35, connector count −0.18, connector stiffness −0.10.

6 Mesh study: how certain are the numbers?

The five best designs were recomputed with identical coils and parameters but a finer mesh: once with 1.5 times as many stations along the coil (same connector length), once with three instead of two cells across the wall thickness. A variant with three wall cells and a fine winding pack needed more than 22 GB of memory and was aborted; instead the winding pack is coarser.

Design	doe1		standard		1.5× stations		wall 3 cells	
	σ_{99}	max.	σ_{99}	max.	σ_{99}	max.	σ_{99}	max.
0008	305	1 695	–	–	407	3 250	306	1 931
0014	377	1 419	–	–	549	3 814	375	1 639
0018	811	2 650	996	4 566	–	–	–	–
0055	366	753	413	942	–	–	–	–
0060	662	1 313	704	1 499	–	–	–	–
0061	351	725	–	–	556	6 553	346	861
0063	369	2 016	595	4 265	–	–	–	–
0084	395	911	462	1 018	–	–	–	–
0093	283	1 190	–	–	410	2 343	280	1 432
0121	264	520	286	749	–	–	–	–
0136	479	995	577	2 238	–	–	–	–
0145	362	1 838	426	2 490	–	–	–	–
0149	837	1 541	636	1 423	–	–	–	–
0169	219	421	308	717	319	1 262	308	794
0176	245	615	434	2 674	–	–	–	–

Table 5: Mesh study: 99-% stress and largest element value in the steel (MPa).

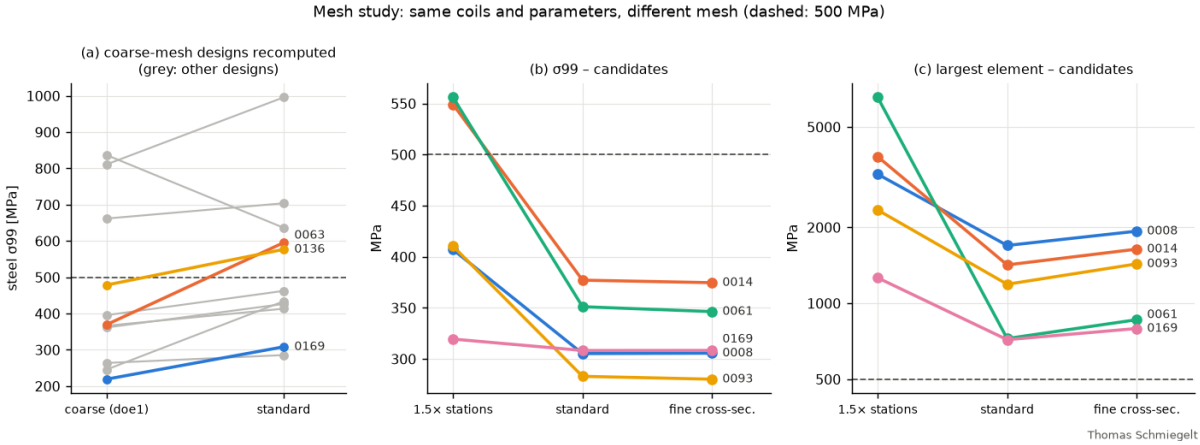


Figure 5: Mesh study of the five best designs.

How to read the figure: Left the 99-% stress, right the largest element value, each for the mesh levels from left to right. A flat line means: the result does not depend on the mesh. On the left four lines rise clearly at the step to 1.5 times as many stations, only 0169 stays almost flat. On the right the maxima grow with every refinement – the signature of a notch or sharp edge where the stress in the model is unbounded.

Where the peaks sit. Of the 200 most highly stressed steel elements, depending on the design 77–100 % lie less than 1 m from a connector joint, only 13–47 % near a bearing. In the model the connectors are sharp-edged steel blocks attached to the side wall of the cases. At their ends along the coil the stiffness jumps – as at the end of a lap joint, where a bond always opens first. Exactly these spots are resolved by the finer mesh along the coil, which is why the stress grows there. For a design this means: the joint needs a shape (fillet, tapered thickness, longer contact area), and the user's idea of weakening areas selectively applies precisely here (section 8).

7 Material and stress categories

7.1 Which steel?

The cases are made of **1.4429** / **X2CrNiMoN17-13-3**, internationally **AISI 316LN**: an austenitic stainless steel with about 17 % chromium, 13 % nickel, 2.5 % molybdenum, very little carbon (L = low carbon, good weldability) and nitrogen (N), which raises the strength and keeps the microstructure stable at low temperature. It is the standard material for magnet cases: ITER builds its field-coil cases from it.

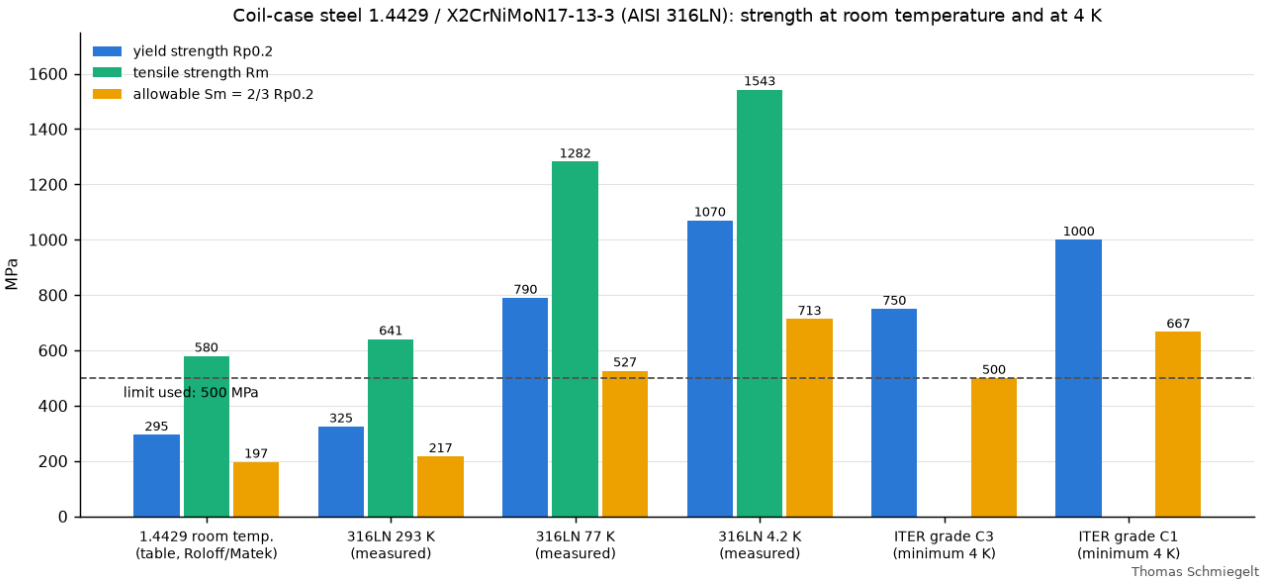


Figure 6: Strength of 1.4429 / 316LN at room temperature and at 4 K, and the allowable stress derived from it.

How to read the figure: Three bars per group: blue the yield strength (beyond it a permanent deformation remains), green the tensile strength (here the specimen breaks), yellow the allowable stress derived from it (two thirds of the yield strength). On the left the handbook value for room temperature, as known from mechanical engineering (295 MPa). Next to it measurements on 316LN: on cooling to 77 K (liquid nitrogen) and 4 K (liquid helium) the steel becomes more than three times as strong. On the right the guaranteed minimum values of the ITER grades. The dashed line is the limit used here.

	$R_{p0.2}$ MPa	R_m MPa	$\frac{2}{3}R_{p0.2}$ MPa	Source
1.4429 solution annealed, room temperature (DIN EN 10088)	295	580	197	Roloff/Matek [1]
316LN-IG, 293 K, measured	325	641	217	[2]
316LN-IG, 77 K, measured	790	1 282	527	[2]
316LN-IG, 4.2 K, measured	1 070	1 543	713	[2]
316LN thick plate, high nitrogen, 4.2 K	≥1 000	–	≥667	[3]
ITER grade C3, 4 K, minimum	750	–	**500**	[4] (unverified)
ITER grade C1, 4 K, minimum	1 000	–	667	[4] (unverified)
ARC assumption, 4 K	1 050	–	700	[5]

Table 6: Properties of 1.4429 / 316LN and the allowable stress derived from them.

Derivation of the 500 MPa. The ITER criteria for magnet structures set the allowable membrane stress to $S_m = \frac{2}{3}R_{p0.2}$ at operating temperature. With the guaranteed minimum of the weakest ITER grade C3 ($R_{p0.2} \geq 750$ MPa at 4 K) this gives 500 MPa. For comparison: the ASME rule $S_m = \min(\frac{2}{3}R_{p0.2}, \frac{1}{3}R_m)$ with the measured 4-K values gives $\min(713, 514) = 514$ MPa – almost the same number. The room-temperature handbook values must not be used for this: at 20 °C only about 200 MPa would be allowed. What matters is that the case is cold in operation; during cool-down and warm-up there is no current and hence no magnetic force.

What the limit does not cover. At 4 K steel fails by crack growth rather than by yielding. Fracture toughness falls with rising strength [3], welds are weaker. ITER therefore additionally checks fatigue and fracture mechanics; both are still missing here.

7.2 Stress categories instead of the 99-% value

For each wall section across the wall (per station, wall and transverse position) the stress tensor was averaged (P_m) and the linear part determined (P_b). Limits with grade C3: $P_m \leq 500$ MPa, $P_m + P_b \leq 750$ MPa.

Model	σ_{99}	P_m max.	P_m 99 %	P_m+P_b max.	peak	connector P_m	location P_m max.
doe1/0061	351	522	379	938	725	189	coil 12, outer wall
doe1/0169	219	359	237	519	421	141	coil 19, side wall
doe1/0093	283	1 147	340	1 233	1 190	168	coil 0, side wall
doe1/0008	305	1 687	344	1 704	1 695	209	coil 7, side wall
doe1/0014	377	1 369	459	1 491	1 419	334	coil 16, side wall
doe1/0017	389	1 631	448	1 813	1 721	131	coil 18, outer wall
doe1/0157	492	1 405	519	1 920	1 658	277	coil 15, side wall
test/0004	481	2 396	736	2 515	2 455	418	coil 12, side wall
netz_normal/0018	996	4 544	1 156	4 589	4 566	358	coil 6, side wall
netz_normal/0055	413	710	411	1 263	942	118	coil 0, side wall
netz_normal/0060	704	1 434	736	1 701	1 499	322	coil 18, outer wall
netz_normal/0063	595	4 261	942	4 270	4 265	259	coil 6, side wall
netz_normal/0084	462	734	455	1 331	1 018	211	coil 14, inner wall
netz_normal/0121	286	675	281	826	749	91	coil 18, side wall
netz_normal/0136	577	2 201	690	2 291	2 238	261	coil 16, side wall
netz_normal/0145	426	2 483	460	2 497	2 490	186	coil 5, side wall
netz_normal/0149	636	1 158	626	1 696	1 423	300	coil 9, outer wall
netz_normal/0169	308	626	333	817	717	201	coil 7, side wall
netz_normal/0176	434	2 664	494	2 685	2 674	296	coil 6, side wall
netz_s15/0008	407	3 230	459	3 270	3 250	221	coil 19, side wall
netz_s15/0014	549	3 747	682	3 882	3 814	286	coil 6, side wall
netz_s15/0061	556	6 551	715	6 555	6 553	272	coil 1, side wall
netz_s15/0093	410	2 320	525	2 366	2 343	164	coil 19, side wall
netz_s15/0169	319	1 258	369	1 265	1 262	196	coil 0, outer wall
netz_q3/0008	306	1 916	352	1 937	1 931	219	coil 19, side wall
netz_q3/0014	375	1 543	481	1 681	1 639	350	coil 16, side wall
netz_q3/0061	346	520	383	957	861	202	coil 12, outer wall
netz_q3/0093	280	1 346	349	1 471	1 432	177	coil 0, side wall
netz_q3/0169	308	631	339	846	794	204	coil 7, side wall
ersatz1/0000	306	717	356	1 209	899	200	coil 6, side wall
ersatz1/0001	220	532	248	746	637	158	coil 12, outer wall
ersatz1/0002	851	7 107	1 541	7 157	7 130	421	coil 1, side wall
ersatz1/0003	272	621	302	724	656	176	coil 14, side wall
ersatz1_s15/0000	385	3 556	481	3 565	3 561	145	coil 20, side wall
ersatz1_s15/0001	366	2 178	469	2 192	2 185	204	coil 19, side wall
ersatz1_s15/0003	394	1 814	482	1 980	1 841	206	coil 12, side wall
schwaechen/0105	358	645	383	895	743	173	coil 7, outer wall
schwaechen/0110	331	635	354	864	734	190	coil 7, side wall
schwaechen/0114	318	631	342	836	725	196	coil 7, side wall
schwaechen/0205	372	1 537	428	1 545	1 541	170	coil 0, outer wall
schwaechen/0210	343	1 394	392	1 402	1 398	185	coil 0, outer wall
schwaechen/0214	330	1 319	378	1 327	1 323	191	coil 0, outer wall

Table 7: Stress categories (MPa). "Peak" = largest element value; connector P_m = averaged over the connector cross-section.

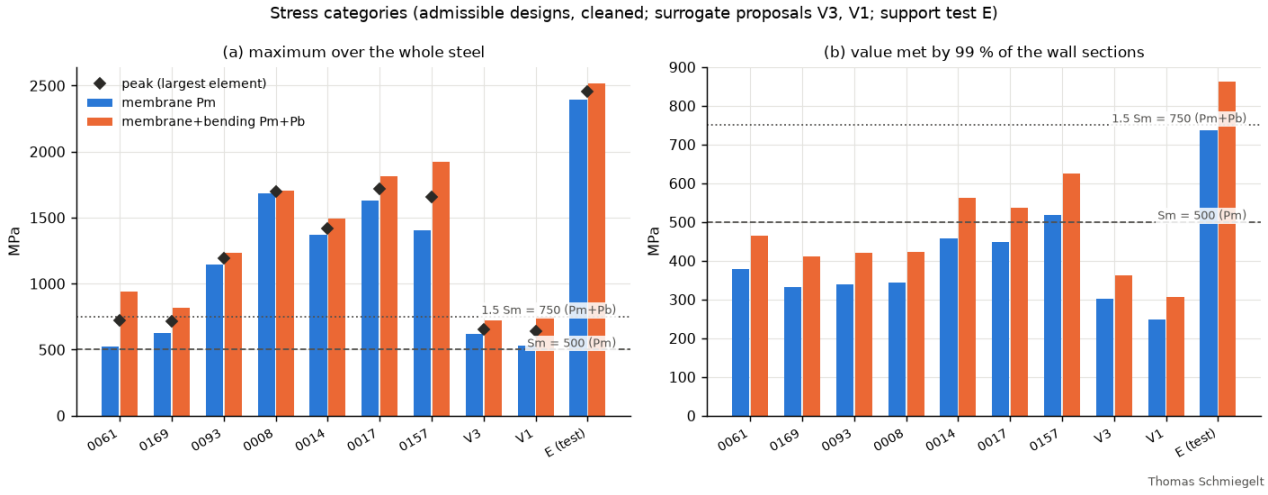


Figure 7: Stress categories: maxima and values in 99 % of the wall sections.

How to read the figure: Two bars per design: membrane P_m (blue) and membrane+bending (orange). Dashed $S_m = 500\,\text{MPa}$ (limit for P_m), dotted $1.5\,S_m$ (limit for $P_m + P_b$). (a) Maximum over the whole steel, diamond = largest single element. For almost all designs the highest membrane stress sits at the same place as the peak, at the end of a connector, where the whole wall thickness is highly loaded: a *local* membrane stress at a discontinuity. No design meets both limits there (0169 was still below on the coarse mesh). (b) Value met by 99 % of the wall sections: 0061, 0169, 0093, 0008, V3 and V1 stay below both limits; 0014, 0017 and 0157 exceed S_m even in the bulk.

Result. For all designs the highest membrane stress lies at the location of the peak, i.e. at the connector joint. Away from these spots (99 % of the wall sections) P_m stays below 400 MPa for the best designs. The assessment "admissible" by the 99-% value is therefore justifiable for the walls; for the connector joints it is not reliable as long as they are modelled sharp-edged.

8 Weakening selectively: softer connectors

Idea and expectation. The user suggested: *"It can also help to weaken certain areas so that the load is distributed more evenly." (translated)* The principle is known from lightweight design: force follows the stiffest path; making an overloaded area more compliant sends load into less stressed areas. Since the peaks sit at the connector ends, the simplest form was tested: all connectors of design 0169 softer (140, 100, 50 GPa instead of 183 GPa), each on the standard mesh and with 1.5 times as many stations. Expected (F-139): peak −30 to −45 %, σ_{99} −5 to −15 %, deformation +20 to +60 %.

No.	E connectors	stations	σ_{99} MPa	max. MPa	deformation mm	strain %	$B \cdot n$ ‰	all limits
0105	50 GPa	64	358	743	27	0.17	2.13	yes
0110	100 GPa	64	331	734	22	0.16	2.13	yes
0114	140 GPa	64	318	725	21	0.15	2.13	yes
0205	50 GPa	96	372	1 541	29	0.19	2.13	yes
0210	100 GPa	96	343	1 398	25	0.18	2.13	yes
0214	140 GPa	96	330	1 323	23	0.17	2.13	yes

Table 8: 0169 with softer connectors. Numbers: 1xx standard mesh, 2xx 1.5 times as many stations; original 183 GPa: 308 MPa and 319 MPa.

Result. The stress in the case rises steadily as the connector stiffness falls: on the standard mesh from 308 MPa (183 GPa) via 318 MPa and 331 MPa to 358 MPa (50 GPa, +16 %); on the fine mesh from 319 MPa to 372 MPa. The maximum rises as well. The deformation grows from 19 mm to 27 mm (+41 %, within the expected range), the winding-pack strain from 0.15 % to 0.17 %. Only the connectors themselves are relieved (99-% value 192 →127 MPa). All variants remain admissible, but with less reserve.

Interpretation. The expectation is refuted, and the reason is instructive: in this structure the connectors are not the cause of the load but the relief path. They shorten the free bending length of the cases. If they become softer overall, the cases bend more again. The peak at the connector end, in contrast, is a consequence of the *shape* of the joint (sharp edge, stiffness jump). The user's idea remains right when applied locally: a connector with soft ends and a stiff core – like a tapered adhesive joint or a chamfered web – introduces the force gradually without giving up the relief path. For this each connector must be split into several bodies with their own stiffness; the stiffnesses can then be iterated by the stress ratio (fully stressed design).

9 Proposals of the surrogate model

Question and expectation. Can the surrogate find better designs by itself? From the six Gaussian processes the smallest reserve was predicted for every conceivable design, and one standard deviation was subtracted (cautious estimate). The four designs with the highest value were fully computed. All four have 24 coils, a coil–plasma distance of 0.21 m, short coils (3.9 m to 4.1 m per coil) and the thickest outer wall; they differ mainly in number and stiffness of the connectors. The cautious prediction was 0.04–0.07. Expected (event log F-142): 1–3 of the 4 admissible, stress ±30 % off the prediction, none clearly more robust than 0169.

Proposal	E_c GPa	c./pair	mass kt	σ_{99} MPa	σ_{99} fine MPa	strain %	$B \cdot n$ ‰	robustness predicted	robustness computed (bottleneck)
V0	55	13	37.8	306	385	0.16	1.52	0.07	0.23 (package space)
V1	77	13	33.1	220	366	0.12	1.87	0.06	0.31 (package space)
V2	81	9	26.4	851	–	0.54	2.19	0.06	-0.79 (winding-pack strain), inadmissible
V3	176	11	29.1	272	394	0.13	1.96	0.04	0.35 (field error B·n)

Table 9: Surrogate proposals. E_c connector stiffness, c./pair connectors per coil pair, "fine" = 1.5 times as many stations. Robustness = smallest reserve over all eight limits; "predicted" = prediction minus one standard deviation.

Result. Three of the four proposals are admissible, and the computed robustness (0.23–0.35) is far above the cautious prediction. V3 (11 stiff connectors per pair, 176 GPa) reaches 0.35 and is, on the standard mesh,

the most robust of all designs computed so far; its bottleneck is the field error, not stress. V2, with only 9 connectors per pair, fails clearly: σ_{99} 851 MPa, winding-pack strain 0.54 %, deformation 177 mm. The surrogate did not see this outlier coming.

Mesh check. As for the doe1 candidates, σ_{99} rises markedly with 1.5 times as many stations: V3 272 → 394 MPa (+45 %), V1 220 → 366 MPa (+66 %), V0 306 → 385 MPa (+26 %). All remain admissible, but V3 drops to a robustness of 0.21 because steel now limits. 0169 (319 MPa on the fine mesh, robustness 0.29) remains the most robust design.

Interpretation. The surrogate is useful for pre-selection: it reliably finds admissible regions (it predicts package, nuclear heating and field error well). For stress it is not enough; a single proposal can be far off, and it does not see the mesh dependence of the peaks at all. Notably, 0169 remains the only design whose stress hardly rises on the finer mesh. In 0169 the connectors merge into almost continuous plates (section 10). A conjecture, not yet tested: fewer free connector ends mean fewer stiffness jumps and hence peaks that do not grow with the mesh.

10 3D models in PicoGK

Why PicoGK. So far 3D models were made in FreeCAD (surfaces, loft, Boolean operations). For twisted stellarator cases, Boolean operations on surfaces often fail because of tiny gaps or self-intersections. PicoGK [6] computes with voxels and distance fields: union, subtraction and offset always succeed, at the cost of a finite resolution.

Construction. Python passes only geometric data to PicoGK, no finished bodies: per station the corner points of the case outer contour, potting outer contour and winding pack (exactly the cross-sections of the computational model), the connector surfaces, the plasma boundary and the bearing locations. In PicoGK (`picogk/StellaratorModell`) these become:

- the coils as closed tubes from the station rectangles, *case = outer contour − potting outer contour*, *potting = potting outer contour − winding pack* (Boolean difference);
- the package by offsetting the plasma: first wall and blanket 0 m to 0.55 m, shield 0.55 m to 0.88 m, vacuum vessel 0.88 m to 1.00 m (assumption, order of magnitude of ARIES-CS);
- the bearings schematically: rods from each bearing patch radially to an inner support cylinder and an outer ring (in the computational model the bearings are rigid; shape and stiffness of the support cylinder are still open);
- three test bodies: package envelope ∩ coils and structure, case ∩ other cases, connectors ∩ winding pack.

Each model is built twice: once finely (40 mm voxels) for volumes and checks, once coarsely (80 mm) for the pictures; the meshes are then thinned. One model takes 2–5 minutes.

Model	voxel mm	time s	WP deviation	case from FE model	connectors	bearings m³	package m³	overlap package/case m³
0004	40	167	−0.9 %	−0.6 %	−1.3 %	1 407	1 442	0.00
0008	40	151	−1.0 %	−0.5 %	−2.1 %	1 469	1 442	0.00
0014	40	137	−0.9 %	−0.5 %	−1.0 %	1 180	1 442	0.00
0017	40	145	−1.0 %	−0.5 %	−0.0 %	1 933	1 442	0.00
0044	40	145	−0.9 %	−0.5 %	−3.6 %	1 660	1 442	0.00
0061	40	150	−1.0 %	−0.5 %	−2.8 %	1 732	1 442	0.00
0063	40	141	−0.9 %	−0.4 %	−3.3 %	887	1 442	0.00
0093	40	197	−1.0 %	−0.5 %	+0.0 %	1 872	1 442	0.00
0103	40	146	−0.9 %	−0.5 %	−1.1 %	2 032	1 442	0.00
0136	40	141	−0.9 %	−0.5 %	−4.1 %	1 706	1 442	0.00
0157	40	139	−0.9 %	−0.5 %	−0.0 %	1 545	1 442	0.00
0169	40	156	−1.0 %	−0.5 %	−12.6 %	1 788	1 442	0.00
0173	40	134	−1.2 %	−0.8 %	−10.3 %	1 090	1 442	0.00
A_test	40	120	−0.7 %	−0.4 %	+0.0 %	757	1 442	0.00
E_test	40	132	−0.7 %	−0.3 %	−0.3 %	1 254	1 442	0.00
V1	40	120	−1.0 %	−0.5 %	−11.3 %	1 917	1 442	0.00
V3	40	137	−1.0 %	−0.6 %	−2.8 %	1 473	1 442	0.00

Table 10: PicoGK models: deviation of the voxel volumes from the computational model and overlaps.

Check. Expected were volume deviations of at most ±3 % and no overlap (F-136). Both are confirmed: winding pack and case lie 0.5–2 % below the computational model (voxels cut off convex edges), the connectors 0–3 % for most designs. **Exception 0169:** there the voxel model lacks 13 % of the connector volume. The cause is not a PicoGK error but a finding: with 13 connectors per coil pair, neighbouring connectors are so close that they touch or interpenetrate; the 312 connectors form only 46 connected bodies. PicoGK unites the bodies and counts every point in space once, while the computational model counts the elements of each connector separately, i.e. the shared space twice. Mass and stiffness of the connectors of 0169 are therefore about 13 % too high in the computational model. Since case stress rises with softer connectors (section 8: 183 → 140 GPa, −24 % stiffness, +3 % σ_{99}), the effect is probably a few percent; the connectors of 0169 should nevertheless be arranged without overlap in a recomputation. The potting (2 cm) is thinner than one voxel and is represented only in part; it is practically invisible in the pictures. No overlap of package and coils confirms the clearance computation of the chain independently, no overlap of the cases the penetration check.

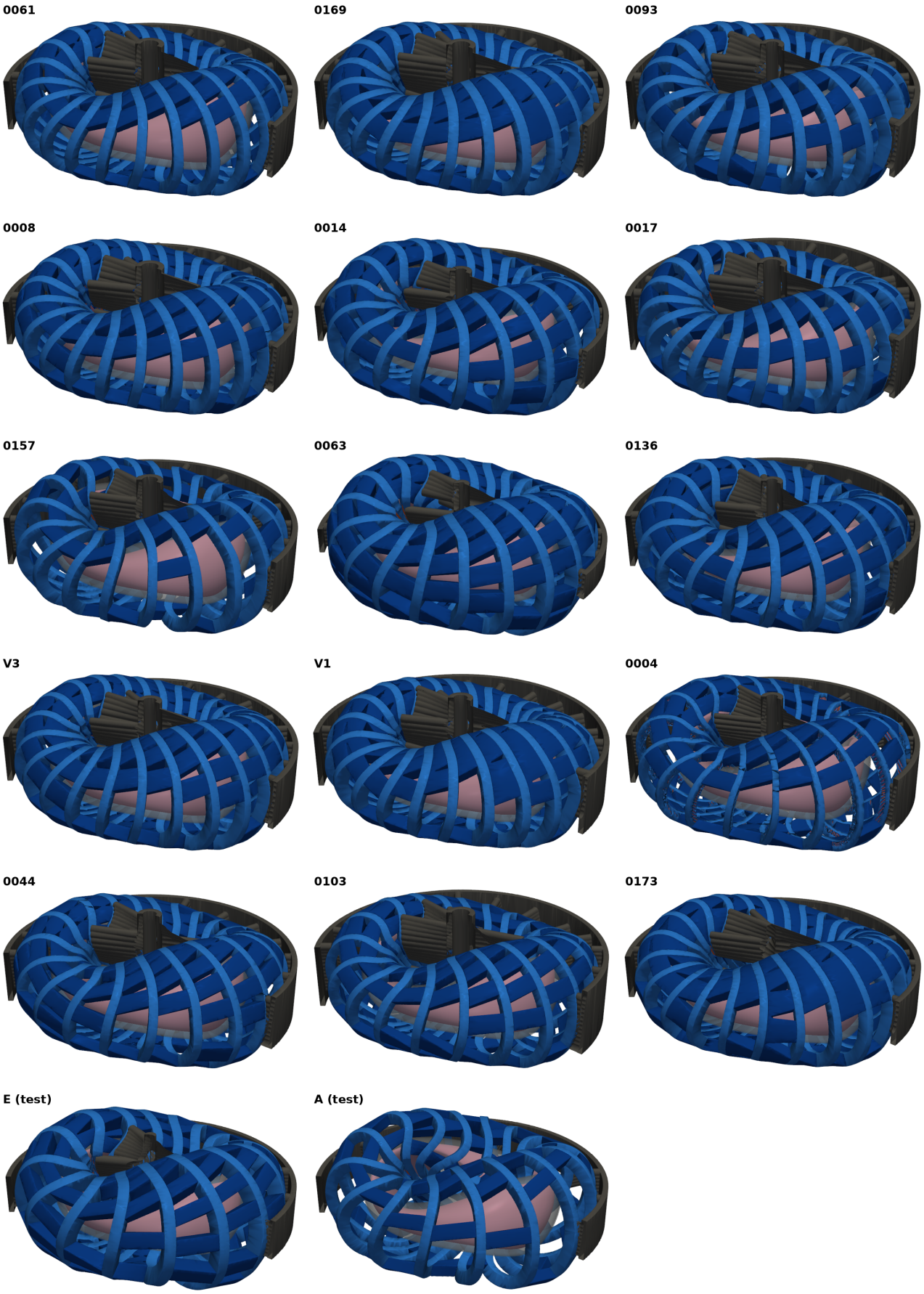


Figure 8: All 3D models of the report (list 1) in overall view.

How to read the figure: Each panel is a complete PicoGK model, labelled with the design number. Blue are

the coil cases, dark blue the connectors, translucent grey the vacuum vessel, pink the plasma, dark grey the schematic bearings. Designs with 24 coils look like a dense cage, the support tests A and E with 16 coils more open; in E the walls are visibly thicker.

11 Data sheets of the designs in list 1

List 1 contains the admissible designs, the five that fail most narrowly, the two that dropped out through the clean-up and the support tests A and E. Each data sheet shows design variables, results, the mesh study, the stress picture and the PicoGK model in a stepped cut.

How to read the figure: Stress picture: top view of all cases, colour from blue (0 MPa) via green and yellow to red (800 MPa and more). Red spots at the sides of the cases are the connector joints. Stepped cut: each layer is cut open at a different angle – from inside to outside plasma (pink), first wall and blanket (yellow), shield (green), vacuum vessel (grey), then the coils with their orange winding pack in cross-section.

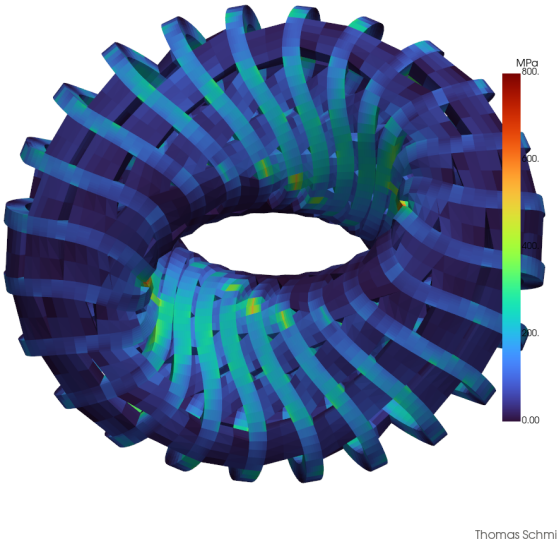
11.1 Design 0061

Verdict: Admissible (cleaned), rank 1 of 7; smallest reserve 30 % (steel σ99).

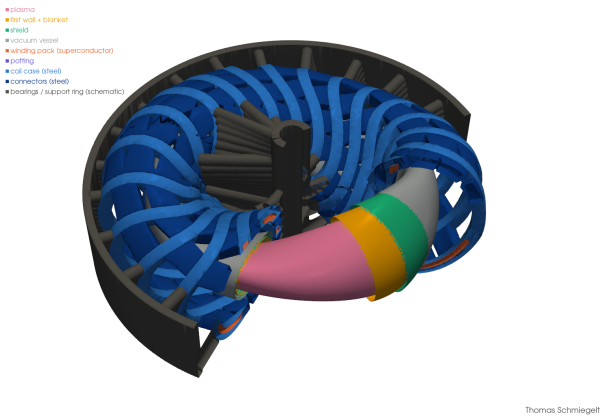
Design variables	
coil–plasma distance	0.204
coils in total	24
coils per half period	6
length per coil	4.58
force weight	0.010
inner wall	0.10
side wall	0.30
outer wall min.	0.52
outer wall extra	0.26
connector stiffness	93
connectors per pair	10
bearing radius	3.23

Results	
σ_{99} steel	351 MPa
largest element	725 MPa
P_m max. / 99 %	522 / 379 MPa
$P_m + P_b$ max.	938 MPa
connectors σ99	191 MPa
max. deformation	18 mm
winding-pack strain	0.17 %
current per coil	14.7 MA
field at conductor	15.6 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	1.49 ‰
clearance plasma–case	1.62 m
nuclear heating / fluence	10 W/m³ / 8.0e16
steel case + connectors	9.5 + 9.9 kt
connectors built	236

Mesh study (σ_{99}/max. in MPa): doe1: 351/725; 1.5× stations: 556/6553; wall 3 cells: 346/861.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 9: Design 0061

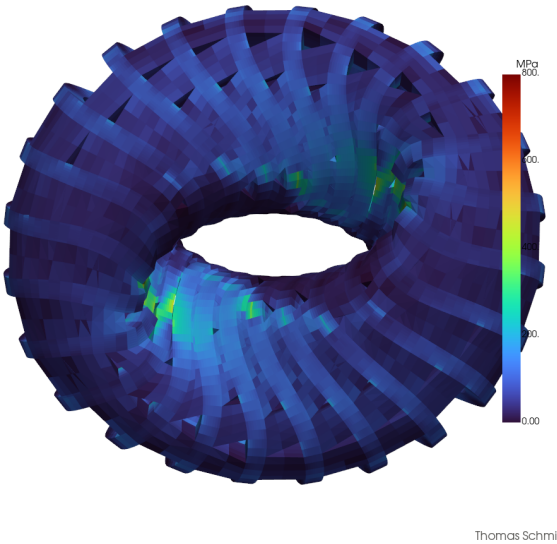
11.2 Design 0169

Verdict: Admissible (cleaned), rank 2 of 7; smallest reserve 29 % (package space).

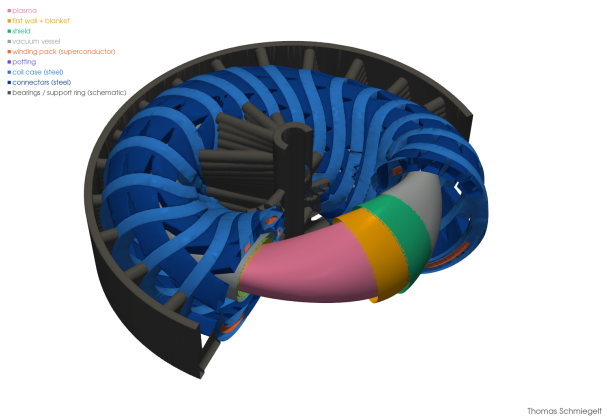
Design variables	
coil–plasma distance	0.197
coils in total	24
coils per half period	6
length per coil	4.87
force weight	0.023
inner wall	0.27
side wall	0.34
outer wall min.	0.67
outer wall extra	0.25
connector stiffness	183
connectors per pair	13
bearing radius	3.29

Results	
σ_{99} steel	308 MPa
largest element	717 MPa
P_m max. / 99 %	626 / 333 MPa
$P_m + P_b$ max.	817 MPa
connectors σ99	192 MPa
max. deformation	19 mm
winding-pack strain	0.15 %
current per coil	14.7 MA
field at conductor	15.5 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	2.13 ‰
clearance plasma–case	1.39 m
nuclear heating / fluence	103 W/m³ / 8.5e17
steel case + connectors	12.8 + 15.2 kt
connectors built	292

Mesh study (σ_{99}/max. in MPa): doe1: 219/421; standard: 308/717; 1.5× stations: 319/1262; wall 3 cells: 308/794.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 10: Design 0169

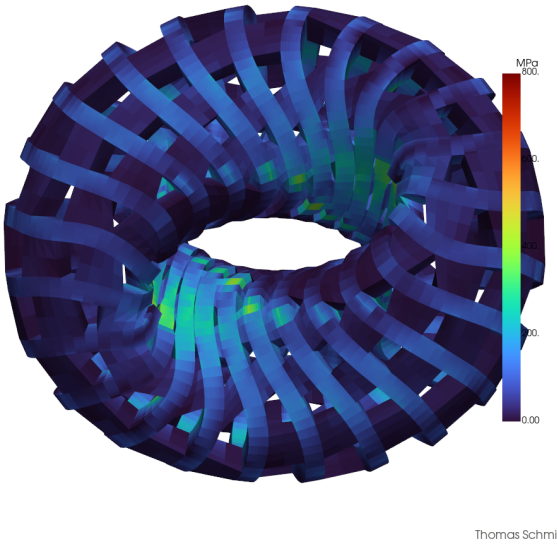
11.3 Design 0093

Verdict: Admissible (cleaned), rank 3 of 7; smallest reserve 27 % (package space).

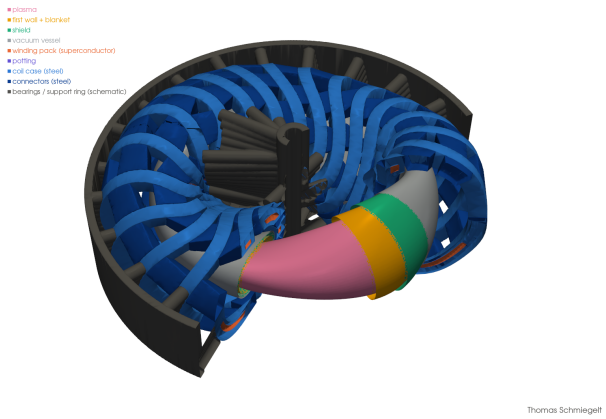
Design variables	
coil–plasma distance	0.206
coils in total	24
coils per half period	6
length per coil	4.14
force weight	0.010
inner wall	0.38
side wall	0.38
outer wall min.	0.73
outer wall extra	0.38
connector stiffness	64
connectors per pair	8
bearing radius	3.49

Results	
σ_{99} steel	283 MPa
largest element	1 190 MPa
P_m max. / 99 %	1 147 / 340 MPa
$P_m + P_b$ max.	1 233 MPa
connectors σ99	154 MPa
max. deformation	20 mm
winding-pack strain	0.16 %
current per coil	14.7 MA
field at conductor	15.7 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	1.45 ‰
clearance plasma–case	1.37 m
nuclear heating / fluence	117 W/m³ / 9.8e17
steel case + connectors	16.5 + 11.6 kt
connectors built	190

Mesh study (σ_{99}/max. in MPa): doe1: 283/1190; 1.5× stations: 410/2343; wall 3 cells: 280/1432.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 11: Design 0093

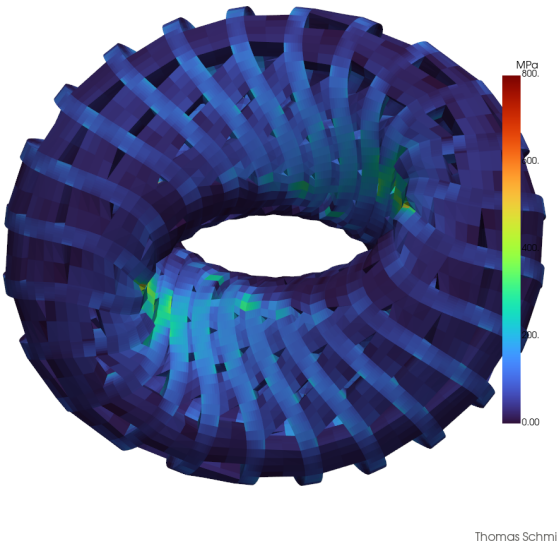
11.4 Design 0008

Verdict: Admissible (cleaned), rank 4 of 7; smallest reserve 27 % (field error B·n).

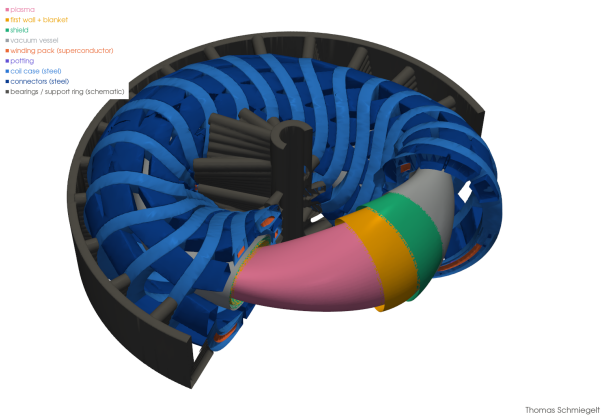
Design variables	
coil–plasma distance	0.211
coils in total	24
coils per half period	6
length per coil	3.66
force weight	0.030
inner wall	0.32
side wall	0.31
outer wall min.	0.72
outer wall extra	0.17
connector stiffness	142
connectors per pair	11
bearing radius	2.80

Results	
σ_{99} steel	305 MPa
largest element	1 695 MPa
P_m max. / 99 %	1 687 / 344 MPa
$P_m + P_b$ max.	1 704 MPa
connectors σ99	187 MPa
max. deformation	19 mm
winding-pack strain	0.17 %
current per coil	14.7 MA
field at conductor	15.7 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	2.20 ‰
clearance plasma–case	1.48 m
nuclear heating / fluence	40 W/m³ / 3.3e17
steel case + connectors	12.2 + 13.7 kt
connectors built	252

Mesh study (σ_{99}/max. in MPa): doe1: 305/1695; 1.5× stations: 407/3250; wall 3 cells: 306/1931.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 12: Design 0008

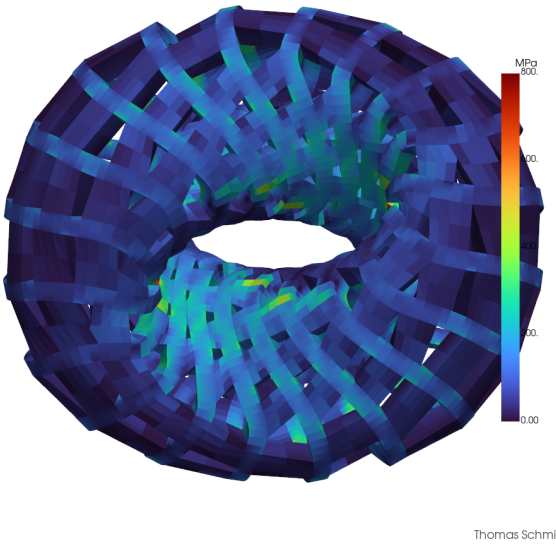
11.5 Design 0014

Verdict: Admissible (cleaned), rank 5 of 7; smallest reserve 25 % (steel σ99).

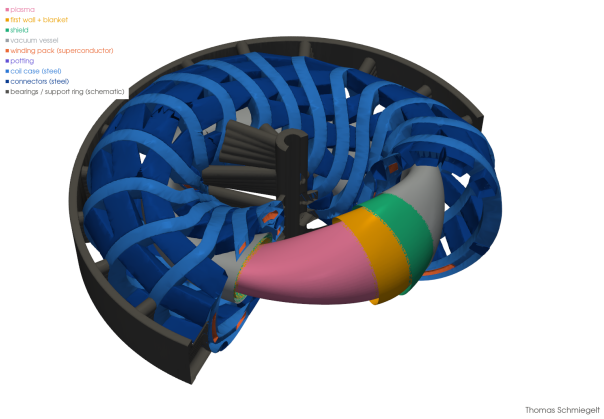
Design variables	
coil–plasma distance	0.233
coils in total	20
coils per half period	5
length per coil	4.34
force weight	0.025
inner wall	0.35
side wall	0.32
outer wall min.	0.63
outer wall extra	0.29
connector stiffness	196
connectors per pair	11
bearing radius	2.04

Results	
σ_{99} steel	377 MPa
largest element	1 419 MPa
P_m max. / 99 %	1 369 / 459 MPa
$P_m + P_b$ max.	1 491 MPa
connectors σ99	243 MPa
max. deformation	18 mm
winding-pack strain	0.21 %
current per coil	17.7 MA
field at conductor	16.9 T
utilisation / margin	32 % / 23 K
field error $B \cdot n$	2.21 ‰
clearance plasma–case	1.65 m
nuclear heating / fluence	8 W/m³ / 6.5e16
steel case + connectors	12.5 + 15.5 kt
connectors built	212

Mesh study (σ_{99}/max. in MPa): doe1: 377/1419; 1.5× stations: 549/3814; wall 3 cells: 375/1639.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

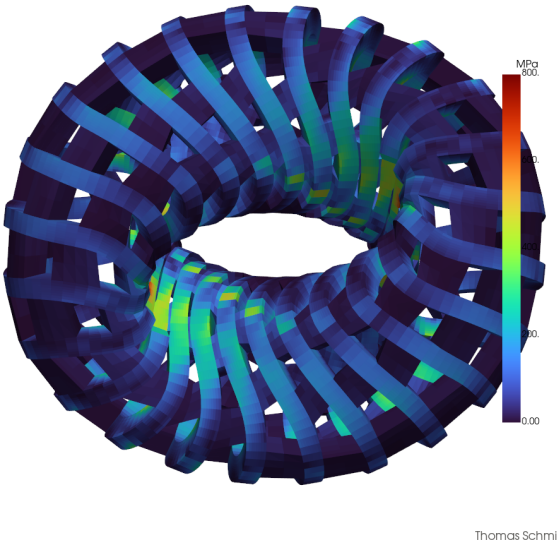
Figure 13: Design 0014

11.6 Design 0017

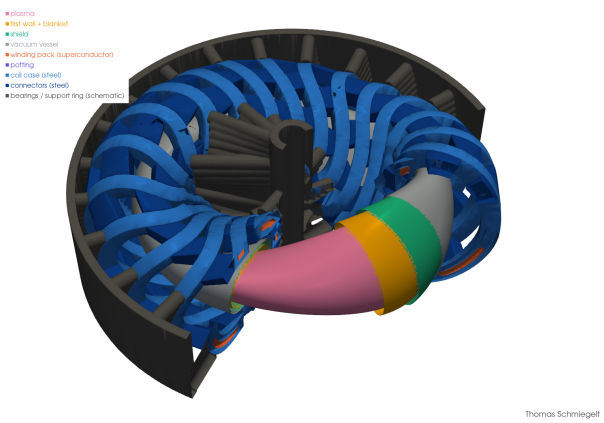
Verdict: Admissible (cleaned), rank 6 of 7; smallest reserve 20 % (package space).

Design variables	
coil–plasma distance	0.220
coils in total	24
coils per half period	6
length per coil	3.77
force weight	0.019
inner wall	0.60
side wall	0.37
outer wall min.	0.43
outer wall extra	0.32
connector stiffness	29
connectors per pair	7
bearing radius	3.47

Results	
σ_{99} steel	389 MPa
largest element	1 721 MPa
P_m max. / 99 %	1 631 / 448 MPa
$P_m + P_b$ max.	1 813 MPa
connectors σ99	131 MPa
max. deformation	31 mm
winding-pack strain	0.20 %
current per coil	14.7 MA
field at conductor	16.0 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	1.66 ‰
clearance plasma–case	1.30 m
nuclear heating / fluence	245 W/m³ / 2.0e18
steel case + connectors	13.9 + 8.7 kt
connectors built	162



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

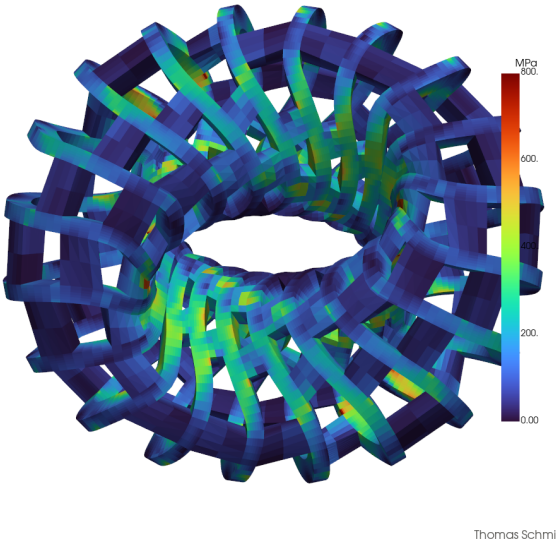
Figure 14: Design 0017

11.7 Design 0157

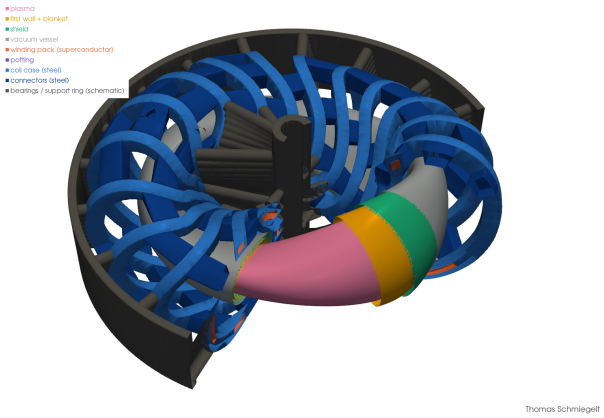
Verdict: Admissible (cleaned), rank 7 of 7; smallest reserve 2 % (steel σ99).

Design variables	
coil–plasma distance	0.218
coils in total	20
coils per half period	5
length per coil	4.09
force weight	0.020
inner wall	0.49
side wall	0.25
outer wall min.	0.58
outer wall extra	0.02
connector stiffness	158
connectors per pair	6
bearing radius	3.06

Results	
σ_{99} steel	492 MPa
largest element	1 658 MPa
P_m max. / 99 %	1 405 / 519 MPa
$P_m + P_b$ max.	1 920 MPa
connectors σ99	335 MPa
max. deformation	34 mm
winding-pack strain	0.22 %
current per coil	17.7 MA
field at conductor	16.5 T
utilisation / margin	31 % / 23 K
field error $B \cdot n$	2.37 ‰
clearance plasma–case	1.35 m
nuclear heating / fluence	142 W/m³ / 1.2e18
steel case + connectors	10.1 + 7.5 kt
connectors built	118



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 15: Design 0157

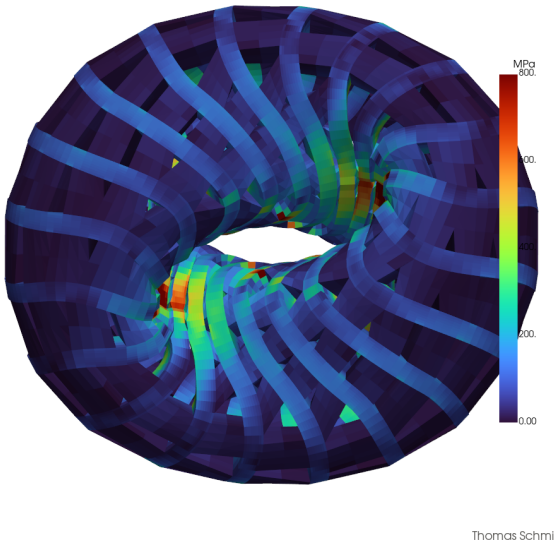
11.8 Design 0063

Verdict: Admissible in doe1, no longer with the standard mesh: bottleneck winding-pack strain (reserve -24 %).

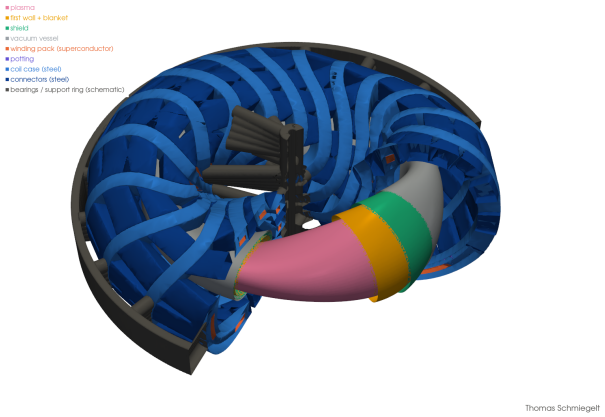
Design variables	
coil–plasma distance	0.177
coils in total	20
coils per half period	5
length per coil	4.45
force weight	0.000
inner wall	0.59
side wall	0.35
outer wall min.	0.69
outer wall extra	0.44
connector stiffness	43
connectors per pair	14
bearing radius	1.71

Results	
σ_{99} steel	595 MPa
largest element	4 265 MPa
P_m max. / 99 %	4 261 / 942 MPa
$P_m + P_b$ max.	4 270 MPa
connectors σ99	200 MPa
max. deformation	81 mm
winding-pack strain	0.37 %
current per coil	17.7 MA
field at conductor	17.5 T
utilisation / margin	33 % / 22 K
field error $B \cdot n$	0.75 ‰
clearance plasma–case	1.53 m
nuclear heating / fluence	24 W/m³ / 2.0e17
steel case + connectors	16.7 + 24.2 kt
connectors built	252

Mesh study (σ_{99}/max. in MPa): doe1: 369/2016; standard: 595/4265.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 16: Design 0063

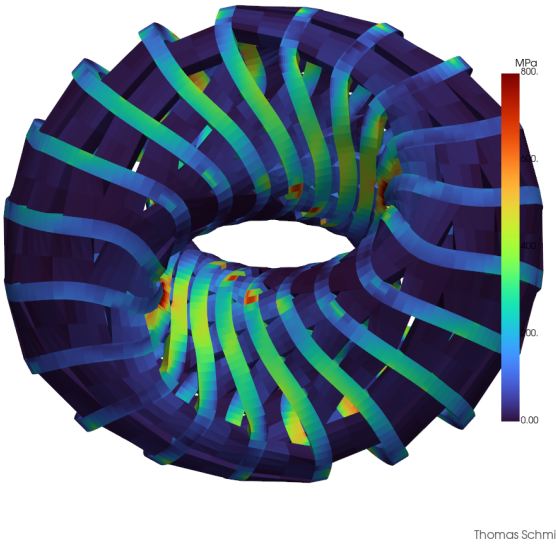
11.9 Design 0136

Verdict: Admissible in doe1, no longer with the standard mesh: bottleneck steel $\sigma 99$ (reserve -15 %).

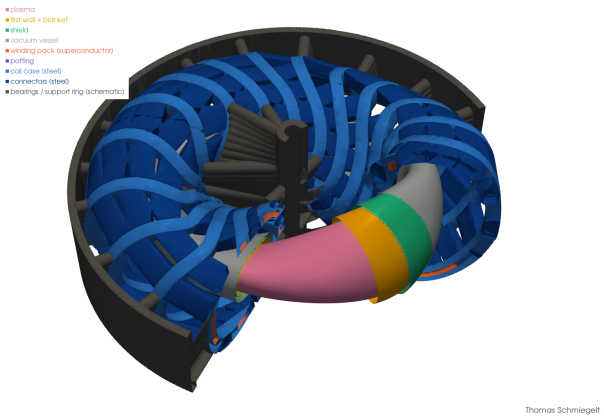
Design variables	
coil–plasma distance	0.232
coils in total	20
coils per half period	5
length per coil	4.51
force weight	0.012
inner wall	0.16
side wall	0.30
outer wall min.	0.57
outer wall extra	0.22
connector stiffness	21
connectors per pair	13
bearing radius	3.12

Results	
σ_{99} steel	577 MPa
largest element	2 238 MPa
P_m max. / 99 %	2 201 / 690 MPa
$P_m + P_b$ max.	2 291 MPa
connectors $\sigma 99$	141 MPa
max. deformation	54 mm
winding-pack strain	0.30 %
current per coil	17.7 MA
field at conductor	16.9 T
utilisation / margin	32 % / 23 K
field error $B \cdot n$	1.62 ‰
clearance plasma–case	1.84 m
nuclear heating / fluence	1 W/m³ / 9.6e15
steel case + connectors	10.0 + 16.3 kt
connectors built	254

Mesh study (σ_{99}/max. in MPa): doe1: 479/995; standard: 577/2238.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 17: Design 0136

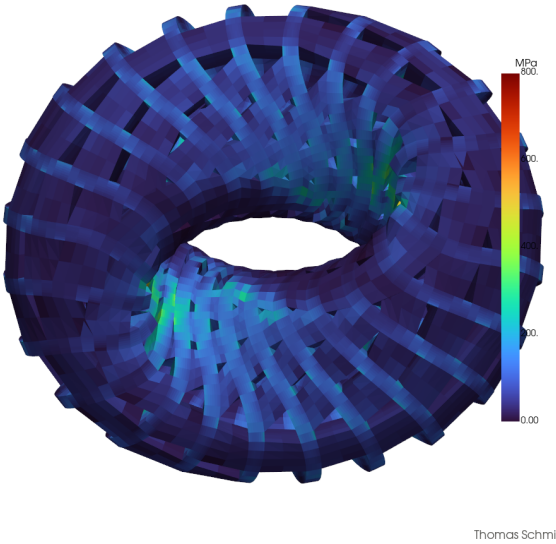
11.10 Surrogate proposal V3

Verdict: Surrogate proposal (section 9), admissible; smallest reserve 35 % (field error B·n).

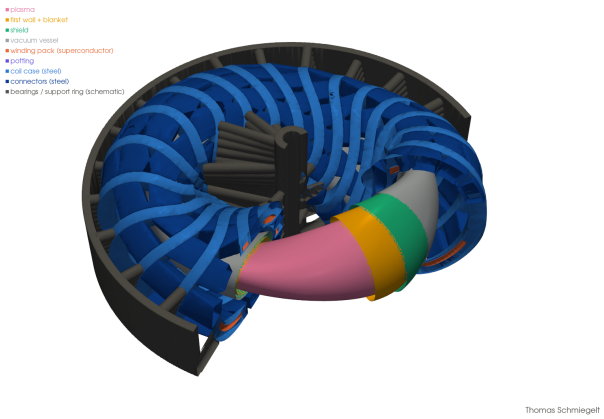
Design variables	
coil–plasma distance	0.205
coils in total	24
coils per half period	6
length per coil	4.15
force weight	0.022
inner wall	0.20
side wall	0.24
outer wall min.	0.77
outer wall extra	0.48
connector stiffness	176
connectors per pair	11
bearing radius	3.02

Results	
σ_{99} steel	272 MPa
largest element	656 MPa
P_m max. / 99 %	621 / 302 MPa
$P_m + P_b$ max.	724 MPa
connectors σ99	178 MPa
max. deformation	13 mm
winding-pack strain	0.13 %
current per coil	14.7 MA
field at conductor	15.6 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	1.96 ‰
clearance plasma–case	1.54 m
nuclear heating / fluence	22 W/m³ / 1.8e17
steel case + connectors	12.3 + 16.8 kt
connectors built	256

Mesh study (σ_{99}/max. in MPa): standard: 272/656; 1.5× stations: 394/1841.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 18: Surrogate proposal V3

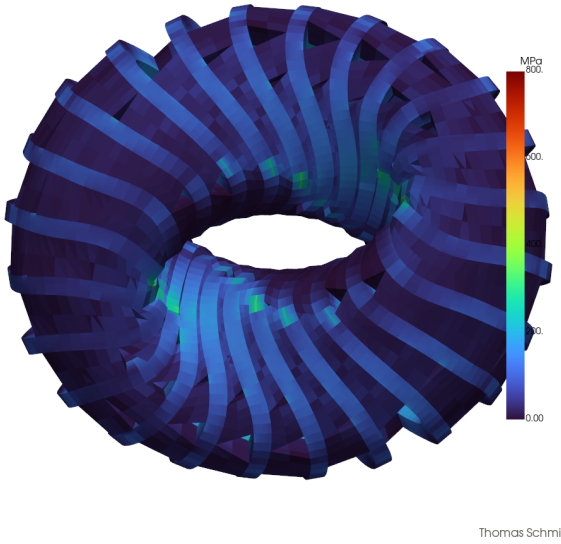
11.11 Surrogate proposal V1

Verdict: Surrogate proposal (section 9), admissible; smallest reserve 31 % (package space).

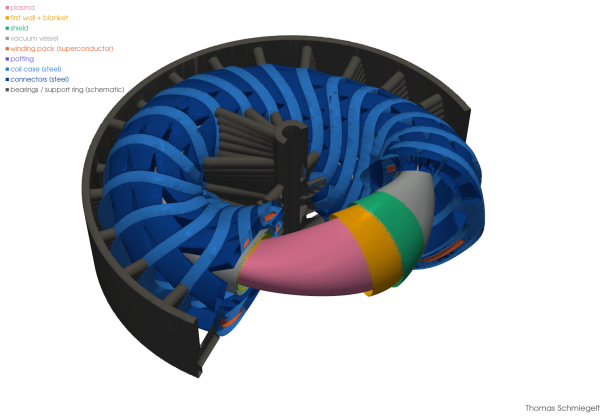
Design variables	
coil–plasma distance	0.208
coils in total	24
coils per half period	6
length per coil	3.98
force weight	0.023
inner wall	0.36
side wall	0.32
outer wall min.	0.77
outer wall extra	0.33
connector stiffness	77
connectors per pair	13
bearing radius	3.41

Results	
σ_{99} steel	220 MPa
largest element	637 MPa
P_m max. / 99 %	532 / 248 MPa
$P_m + P_b$ max.	746 MPa
connectors σ99	111 MPa
max. deformation	11 mm
winding-pack strain	0.12 %
current per coil	14.7 MA
field at conductor	15.6 T
utilisation / margin	30 % / 24 K
field error $B \cdot n$	1.87 ‰
clearance plasma–case	1.41 m
nuclear heating / fluence	85 W/m³ / 7.1e17
steel case + connectors	14.6 + 18.5 kt
connectors built	308

Mesh study (σ_{99}/max. in MPa): standard: 220/637; 1.5× stations: 366/2185.



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

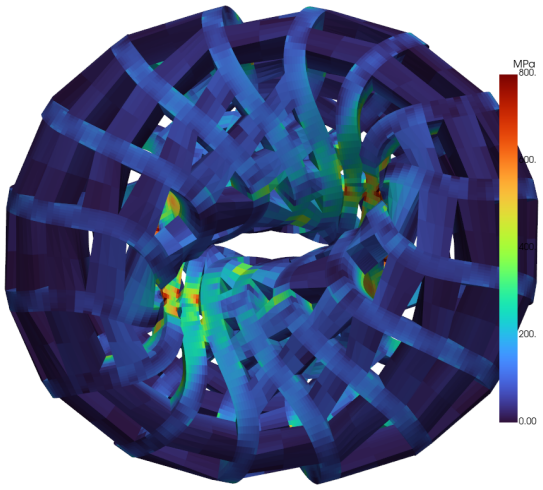
Figure 19: Surrogate proposal V1

11.12 Support test E (thickest walls)

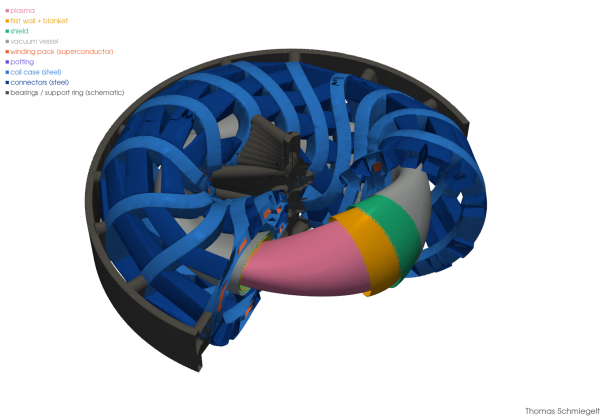
Verdict: Same 16 coils as the starting design; only the support structure differs.

Design variables	
coil–plasma distance	0.180
coils in total	16
coils per half period	4
length per coil	4.50
force weight	0.000
inner wall	0.60
side wall	0.40
outer wall min.	0.80
outer wall extra	0.60
connector stiffness	200
connectors per pair	12
bearing radius	2.50

Results	
σ_{99} steel	481 MPa
largest element	2 455 MPa
P_m max. / 99 %	2 396 / 736 MPa
$P_m + P_b$ max.	2 515 MPa
connectors σ99	276 MPa
max. deformation	41 mm
winding-pack strain	0.31 %
current per coil	22.1 MA
field at conductor	18.7 T
utilisation / margin	34 % / 22 K
field error $B \cdot n$	2.71 ‰
clearance plasma–case	1.53 m
nuclear heating / fluence	25 W/m³ / 2.1e17
steel case + connectors	17.6 + 19.4 kt
connectors built	168



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

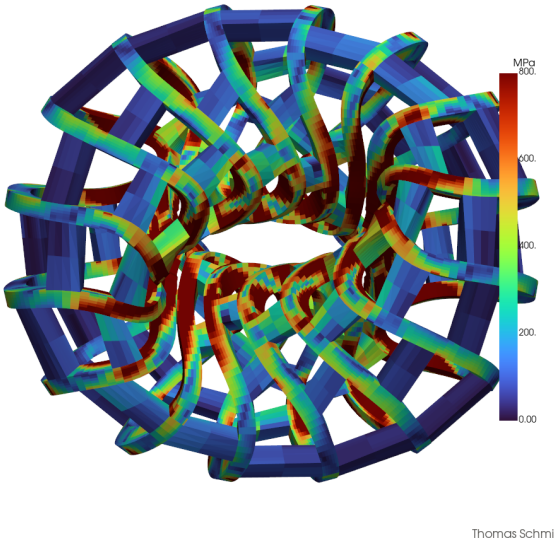
Figure 20: Support test E (thickest walls)

11.13 Support test A (starting design)

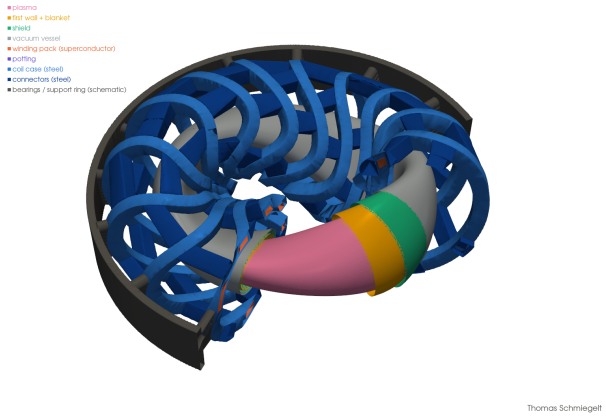
Verdict: Same 16 coils as the starting design; only the support structure differs.

Design variables	
coil–plasma distance	0.180
coils in total	16
coils per half period	4
length per coil	4.50
force weight	0.000
inner wall	0.35
side wall	0.24
outer wall min.	0.45
outer wall extra	0.30
connector stiffness	50
connectors per pair	6
bearing radius	2.50

Results	
σ_{99} steel	2 129 MPa
largest element	8 454 MPa
connectors σ99	675 MPa
max. deformation	399 mm
winding-pack strain	0.76 %
current per coil	22.1 MA
field at conductor	18.7 T
utilisation / margin	34 % / 22 K
field error $B \cdot n$	2.71 ‰
clearance plasma–case	1.77 m
nuclear heating / fluence	2 W/m³ / 1.8e16
steel case + connectors	8.5 + 7.0 kt
connectors built	90



von Mises stress in the steel (top view, 0–800 MPa)



PicoGK model, stepped cut

Figure 21: Support test A (starting design)

12 Comparison and conclusions

Which design is the most robust? At the user's request the goal was not the lightest design but the one with the largest reserve against all eight limits ("weight is not that important, it is a building"). After the clean-up seven designs meet all limits. Their smallest reserve (robustness) ranges from 0.30 (0061) to 0.02 (0157). Two designs are almost level: 0061 (bottleneck steel) and 0169 (bottleneck package thickness). The mesh study decides (section 6):

- **0169** stays practically unchanged on a finer mesh (308 → 319 MPa, +4 %). Its reserve does not depend on numerical accuracy; its bottleneck is the package thickness, which depends on geometry, not on the mesh. It is therefore the **recommended design**: 24 coils, 13 stiff connectors per coil pair, about 28 kt of steel.
- **0061** has the largest reserve on the standard mesh but rises by +58 % on the fine mesh (351 → 556 MPa) and then exceeds the limit. Its bottleneck is exactly the mesh-dependent quantity.
- **0093, 0008, 0014** rise by 34–46 % on the fine mesh and end up partly just below, partly above the limit. They are candidates, but only after a more accurate analysis of the connector joints.
- **0017, 0157** have little reserve and were not refined further.

- **V3, V1** (surrogate proposals) are more robust than 0169 on the standard mesh (0.35 and 0.31), but on the fine mesh their stress rises by 45 and 66 % (section 9). V3 thus drops to 0.21.

Narrow misses. The five most robust inadmissible designs (0173, 0103, 0044, 0004, 0136) fail almost all on steel stress or on the field error B_n. 0063 and 0136 were admissible on the coarse mesh and dropped out after recomputation – an example of how easily an unnoticed change of mesh flatters the result.

What distinguishes the variants. Across all 198 designs:

1. **Number of coils:** no design with 16 coils was admissible. The coils become so long and so strongly curved that the cases bend too much, and the field error is larger. 20 and 24 coils split the force over more and shorter members.
2. **Coil shape before structure:** coil–plasma distance and coil length determine the stress more than number or stiffness of the connectors. The structure can compensate an unfavourable coil shape only to a limited extent.
3. **Package and nuclear heating** are almost purely geometric and are predicted very well by the surrogate ($R^2 \approx 0.98$). They can be screened reliably before any strength analysis.
4. **Stress** is predicted only roughly by the surrogate ($R^2 \approx 0.46$) because it depends on local peaks at the connector joints. Here the FE analysis remains indispensable.
5. **Softer connectors** (user idea "weaken selectively", first form) increase the stress. The idea has to be applied locally: soft ends, stiff core.

How reliable is this? The numbers come from a simplified model: cases as rectangular cross-sections, connectors as beam bundles, supports as fixed points, only the magnetic force in cold operation (no cool-down stress, no dead weight, no fatigue). The finer mesh shows that the peaks are not yet converged. The statement "0169 is robust, 16 coils do not work" is reliable because it holds on all meshes; the exact megapascals are not yet.

Next steps.

1. Compute 0169 with locally graded connectors (soft ends) and rounded joints; adapt stiffnesses iteratively by the stress ratio.
2. Test the conjecture that 0169 is mesh-stable because its connectors merge into almost continuous plates: compute 0169 once with non-overlapping connectors and once with a deliberately continuous plate.
3. Check mesh convergence at the connector joints of 0169 and 0093 with targeted local refinement.
4. Add cool-down stress (293 K → 4 K) and dead weight.
5. Reuse the PicoGK model of 0169 as the starting shape for supports, cooling channels and potting.

A All designs of the study doe1

Abbreviations: dist. minimum coil–plasma distance (unit geometry), length per coil (unit geometry), force = force weight, w.i./w.s. inner/side wall, t outer wall min., n_c connectors per pair, E_c connector stiffness, R_b bearing radius, clear. = clearance plasma–case, heat = nuclear heating of the winding pack. Last column: ✓ admissible, otherwise bottleneck. * = recomputed with the standard mesh.

No.	coils	dist. –	length –	force –	w.i. m	w.s. m	t m	n_c	E_c GPa	R_b m	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clear. m	heat W/m³	steel kt	ok
0000	24	0.18	4.4	0.024	0.10	0.35	0.58	10	70	2.5	840	3 425	0.42	1.98	1.44	60	23	stahl
0001	20	0.24	4.4	0.012	0.13	0.15	0.35	7	135	2.9	1 456	3 851	0.60	1.53	2.00	0	12	stahl
0002	24	0.30	4.5	0.028	0.26	0.10	0.20	8	25	2.7	4 721	13 357	2.01	3.67	2.39	0	13	stahl
0003	16	0.20	3.7	0.025	0.34	0.10	0.49	12	142	1.9	617	1 361	0.26	9.48	1.40	90	20	Bn
0004	20	0.19	4.6	0.012	0.12	0.08	0.50	10	148	3.2	558	1 085	0.26	1.81	1.54	22	19	stahl
0005	16	0.15	4.9	0.020	0.59	0.08	0.15	13	114	2.9	844	1 847	0.35	2.40	1.26	372	23	stahl
0006	24	0.12	4.4	0.012	0.50	0.14	0.18	6	78	2.9	536	1 222	0.22	1.77	0.99	5 689	14	fluenz
0007	16	0.16	3.7	0.019	0.14	0.38	0.36	8	88	2.6	870	3 574	0.42	9.92	1.60	12	17	Bn
0008	24	0.21	3.7	0.030	0.32	0.31	0.72	11	142	2.8	305	1 695	0.17	2.20	1.48	40	26	✓
0009	24	0.23	4.1	0.008	0.49	0.25	0.28	13	145	3.0	1 159	9 216	0.62	1.50	1.55	19	24	stahl
0010	16	0.27	4.6	0.023	0.41	0.20	0.46	8	149	1.4	1 381	8 739	0.63	2.80	1.93	0	21	stahl
0011	16	0.18	4.9	0.023	0.12	0.09	0.45	7	42	1.6	3 096	19 166	1.50	3.25	1.80	2	14	stahl

No.	coils	dist. –	length –	force –	w.i. m	w.s. m	t m	n_c	E_c GPa	R_b m	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clear. m	heat W/m³	steel kt	ok
0012	16	0.21	3.9	0.009	0.20	0.17	0.45	9	50	2.7	999	2 952	0.40	7.17	1.68	5	13	Bn
0013	16	0.13	4.3	0.013	0.24	0.14	0.77	14	103	2.4	773	1 718	0.34	4.32	1.63	9	25	stahl
0014	20	0.23	4.3	0.025	0.35	0.32	0.63	11	196	2.0	377	1 419	0.21	2.21	1.65	8	28	✓
0015	24	0.29	4.4	0.002	0.14	0.16	0.65	7	168	3.8	3 586	30 683	2.18	1.12	2.48	0	18	dehn
0016	20	0.29	4.9	0.017	0.16	0.20	0.71	8	172	2.5	3 499	15 129	1.95	2.24	2.42	0	19	stahl
0017	24	0.22	3.8	0.019	0.60	0.37	0.43	7	29	3.5	389	1 721	0.20	1.66	1.30	245	23	✓
0018*	20	0.25	4.4	0.010	0.14	0.13	0.55	14	26	1.0	996	4 566	0.44	1.43	2.08	0	24	stahl
0019	16	0.25	3.7	0.022	0.18	0.30	0.76	9	24	3.1	1 571	14 989	0.96	13.15	1.93	0	17	Bn
0020	16	0.24	4.8	0.001	0.28	0.11	0.27	14	137	3.3	1 363	6 952	0.55	3.21	1.78	2	17	stahl
0021	24	0.16	4.3	0.026	0.58	0.23	0.53	8	138	3.8	558	4 173	0.29	2.26	0.87	18 176	26	fluenz
0022	20	0.23	4.5	0.000	0.44	0.16	0.56	8	37	3.9	600	2 183	0.26	0.70	1.57	17	24	stahl
0023	20	0.28	4.8	0.027	0.39	0.38	0.22	6	156	3.9	1 864	10 858	1.13	2.89	2.06	0	15	dehn
0024	20	0.26	4.7	0.010	0.45	0.11	0.52	11	24	1.1	1 787	7 614	0.85	1.55	1.82	1	24	stahl
0025	16	0.18	4.7	0.000	0.26	0.34	0.45	14	41	3.7	1 010	9 842	0.50	1.94	1.88	1	26	stahl
0026	20	0.19	4.5	0.003	0.51	0.18	0.68	11	79	1.9	809	2 253	0.38	1.43	1.41	82	29	stahl
0027	20	0.28	4.7	0.004	0.31	0.21	0.16	10	171	3.8	2 080	10 143	1.02	2.25	2.18	0	16	stahl
0028	20	0.17	4.9	0.008	0.50	0.29	0.74	9	22	3.7	888	3 682	0.40	1.45	1.33	178	30	stahl
0029	16	0.29	3.9	0.020	0.15	0.36	0.64	13	27	2.9	1 975	34 584	1.38	10.34	2.35	0	22	dehn
0030	24	0.28	4.6	0.027	0.37	0.15	0.54	13	71	1.1	1 960	16 426	1.07	2.89	2.06	0	31	stahl
0031	24	0.24	3.9	0.016	0.23	0.28	0.66	8	190	1.0	1 514	8 426	0.97	1.99	1.84	1	21	dehn
0032	16	0.17	4.0	0.003	0.51	0.31	0.63	12	38	1.3	777	3 008	0.28	6.70	1.48	39	25	Bn
0033	24	0.25	4.1	0.025	0.59	0.27	0.73	8	118	2.4	1 019	4 409	0.68	1.78	1.56	18	25	dehn
0034	16	0.26	4.5	0.020	0.33	0.32	0.60	6	21	2.6	1 574	4 775	0.56	3.08	1.96	0	17	stahl
0035	24	0.18	4.7	0.006	0.31	0.10	0.62	12	50	1.9	1 892	5 956	0.83	1.23	1.31	231	30	stahl
0036	20	0.25	3.8	0.021	0.44	0.22	0.20	7	119	1.8	683	7 457	0.38	3.45	1.74	3	15	stahl
0037	24	0.13	4.9	0.009	0.14	0.19	0.32	6	53	1.6	1 800	4 950	0.66	1.33	1.43	65	12	stahl
0038	20	0.17	3.9	0.017	0.47	0.22	0.67	13	60	2.1	525	2 561	0.23	3.27	1.15	1 095	32	fluenz
0039	16	0.28	4.2	0.001	0.41	0.27	0.16	7	31	3.0	1 548	6 664	0.70	4.38	2.05	0	14	stahl
0040	24	0.29	3.9	0.010	0.12	0.22	0.71	12	91	2.6	1 262	9 182	1.02	3.84	2.41	0	23	dehn
0041	16	0.14	4.2	0.023	0.21	0.21	0.22	9	73	3.9	843	1 629	0.36	4.96	1.60	12	13	stahl
0042	20	0.14	3.9	0.002	0.19	0.35	0.50	12	170	2.0	632	3 962	0.34	2.66	1.61	11	21	stahl
0043	24	0.17	3.7	0.008	0.22	0.08	0.48	9	40	1.5	653	1 208	0.29	1.79	1.30	244	16	stahl
0044	20	0.16	4.7	0.029	0.23	0.26	0.30	13	110	3.2	540	1 583	0.23	2.76	1.35	151	26	stahl
0045	20	0.15	3.7	0.028	0.21	0.13	0.75	9	175	2.0	600	3 060	0.29	4.74	1.36	136	22	Bn
0046	16	0.16	4.0	0.021	0.38	0.27	0.62	9	46	3.3	619	2 502	0.26	6.36	1.43	68	22	Bn
0047	24	0.15	4.5	0.011	0.46	0.23	0.43	8	34	2.2	983	4 736	0.42	1.48	1.10	1 825	25	fluenz
0048	16	0.29	3.9	0.016	0.23	0.39	0.46	7	160	3.6	2 081	19 844	1.39	10.18	2.32	0	13	dehn
0049	16	0.21	4.9	0.015	0.27	0.20	0.31	9	84	3.2	2 808	9 212	1.07	2.42	1.74	3	18	stahl
0051	20	0.17	3.8	0.009	0.17	0.33	0.32	10	37	1.9	1 152	2 169	0.50	3.32	1.50	32	15	stahl
0052	24	0.19	4.1	0.017	0.17	0.31	0.39	13	55	2.2	1 321	4 323	0.52	2.06	1.39	95	20	stahl
0053	20	0.25	4.2	0.013	0.20	0.26	0.19	6	57	2.0	1 314	9 639	0.57	1.71	2.02	0	11	stahl
0054	16	0.16	3.9	0.011	0.41	0.16	0.17	7	62	2.2	963	2 479	0.37	7.15	1.42	72	12	Bn
0055*	20	0.13	4.0	0.027	0.43	0.17	0.72	14	38	2.9	413	942	0.18	2.84	1.15	1 052	32	fluenz
0056	20	0.28	5.0	0.025	0.44	0.13	0.21	8	59	2.8	2 615	13 038	1.21	2.25	2.06	0	16	stahl
0057	20	0.24	4.4	0.025	0.43	0.15	0.23	6	31	1.1	1 056	4 063	0.39	2.18	1.60	12	14	stahl
0058	16	0.23	3.6	0.015	0.34	0.38	0.78	10	32	1.2	689	2 099	0.29	6.71	1.60	12	29	Bn
0059	16	0.19	4.6	0.002	0.58	0.14	0.74	11	110	2.8	1 161	4 304	0.57	2.27	1.48	41	31	stahl
0060*	24	0.22	3.7	0.014	0.28	0.26	0.19	13	155	3.5	704	1 499	0.29	1.61	1.66	7	16	stahl
0061	24	0.20	4.6	0.010	0.10	0.30	0.52	10	93	3.2	351	725	0.17	1.49	1.62	10	19	✓
0062	24	0.30	4.7	0.030	0.26	0.26	0.30	9	21	2.1	2 627	13 807	1.58	3.27	2.40	0	17	dehn
0063*	20	0.18	4.5	0.000	0.59	0.35	0.69	14	43	1.7	595	4 265	0.37	0.75	1.53	24	41	dehn
0064	16	0.27	4.1	0.022	0.50	0.22	0.28	12	131	2.4	683	5 601	0.35	5.70	1.85	1	22	Bn
0065	20	0.17	4.1	0.007	0.11	0.12	0.44	9	48	4.0	660	1 210	0.31	1.97	1.61	11	15	stahl
0066	24	0.14	4.5	0.012	0.12	0.18	0.40	7	112	2.5	3 162	8 139	1.28	1.92	1.40	91	14	stahl
0067	20	0.20	4.2	0.018	0.18	0.37	0.73	11	82	3.5	999	2 826	0.47	3.08	1.53	24	28	stahl
0068	20	0.12	4.2	0.006	0.22	0.14	0.24	10	75	1.3	1 012	2 393	0.42	1.57	1.53	25	18	stahl
0069	20	0.29	3.8	0.017	0.37	0.20	0.50	7	130	3.5	1 624	16 704	0.84	6.85	2.16	0	15	stahl
0070	24	0.18	4.8	0.003	0.29	0.19	0.42	14	119	2.4	1 172	6 574	0.56	0.83	1.51	31	28	stahl
0071	24	0.25	4.4	0.030	0.57	0.39	0.38	12	192	1.2	1 075	5 955	0.66	1.55	1.61	11	30	dehn
0072	24	0.17	4.3	0.022	0.16	0.34	0.74	9	95	2.3	1 093	8 828	0.56	2.35	1.33	188	22	stahl
0073	16	0.24	3.6	0.001	0.46	0.14	0.78	12	52	3.4	1 030	50 101	0.71	13.89	1.54	23	23	Bn
0074	16	0.26	4.8	0.028	0.52	0.15	0.34	6	85	3.1	1 715	23 336	0.65	2.71	1.74	3	14	stahl
0075	20	0.20	3.7	0.012	0.48	0.11	0.27	10	163	1.4	477	1 064	0.20	4.62	1.25	416	18	Bn
0076	20	0.27	4.9	0.016	0.24	0.32	0.59	9	33	1.7	1 983	10 425	1.35	2.07	2.13	0	23	dehn
0077	24	0.21	4.8	0.007	0.18	0.18	0.23	10	47	1.5	779	1 288	0.31	1.07	1.65	7	14	stahl
0078	24	0.24	4.6	0.024	0.27	0.11	0.37	8	36	4.0	1 919	14 222	0.85	1.52	1.85	1	18	stahl
0079	20	0.27	4.8	0.002	0.36	0.19	0.25	13	30	3.7	1 320	8 604	0.66	1.17	2.00	0	23	stahl
0080	16	0.13	4.3	0.015	0.40	0.13	0.29	12	65	2.6	803	1 759	0.32	3.98	1.47	45	21	stahl
0081	24	0.27	3.7	0.024	0.17	0.35	0.79	9	97	3.8	577	5 361	0.41	2.32	2.17	0	19	dehn
0082	20	0.16	4.6	0.006	0.19	0.29	0.33	8	23	1.2	2 591	34 831	1.50	1.75	1.72	4	17	stahl
0083	16	0.26	4.1	0.012	0.15	0.24	0.59	6	27	3.6	996	5 784	0.54	5.30	2.05	0	14	stahl
0084*	16	0.19	3.7	0.015	0.50	0.18	0.35	14	63	3.9	462	1 018	0.20	9.75	1.26	372	20	Bn
0085	20	0.19	4.4	0.013	0.49	0.25	0.72	10	22	2.8	608	2 067	0.28	2.00	1.23	473	30	fluenz
0086	16	0.20	4.7	0.023	0.19	0.36	0.25	13	67	2.7	1 339	5 294	0.53	2.87	1.64	8	21	stahl
0087	20	0.21	4.7	0.006	0.52	0.19	0.24	6	30	1.7	1 837	11 946	0.94	1.40	1.32	198	17	stahl
0088	24	0.14	4.9	0.017	0.56	0.40	0.39	13	86	1.4	1 478	5 859	0.71	1.92	0.91	12 191	33	fluenz
0089	24	0.15	4.0	0.014	0.48	0.34	0.67	12	22	2.7	645	2 827	0.30	2.16	1.07	2 325	30	fluenz
0090	20	0.19	4.5	0.020	0.35	0.25	0.51	6	161	1.2	1 929	10 870	0.91	2.84	1.36	136	17	stahl
0091	16	0.22	4.3	0.024	0.36	0.22	0.48	10	42	2.5	848	2 510	0.38	4.47	1.52	28	23	stahl
0092	20	0.23	4.2	0.029	0.49	0.38	0.53	10	44	1.3	1 195	14 134	0.75	3.37	1.43	67	23	dehn
0093	24	0.21	4.1	0.010	0.38	0.38	0.73	8	64	3.5	283	1 190	0.16	1.45	1.37	117	28	✓
0094	24	0.15	4.0	0.027	0.29	0.23	0.64	11	178	3.7	597	2 665	0.30	2.77	1.17	887	26	fluenz
0095	16	0.19	4.6	0.026	0.55	0.10	0.51	8	186	3.6	773	2 138	0.32	3.25	1.28	311	22	stahl
0096	16	0.25	3.8	0.010	0.29	0.12	0.25	11	25	3.0	1 006	3 209	0.40	8.54	1.85	1	15	Bn
0097	24	0.14	4.7	0.029	0.19	0.29	0.25	7	99	3.8	2 155	28 026	0.95	2.76	1.30	251	14	stahl
0098	20	0.29	3.7	0.006	0.36	0.16	0.41	13	25	1.8	2 250	13 304	1.30	6.96	2.18	0	20	stahl
0099	20	0.28	4.9	0.027	0.56	0.15	0.26	8	62	1.6	2 025	7 095	0.97	2.32	1.89	1	20	stahl
0100	24	0.13	3.7	0.020	0.24	0.09	0.52	7	59	3.0	441	788	0.20	2.37	1.19	730	19	fluenz

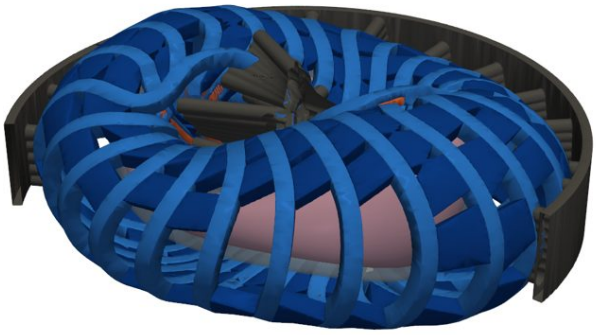
No.	coils	dist. –	length –	force –	w.i. m	w.s. m	t m	n_c	E_c GPa	R_b m	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clear. m	heat W/m³	steel kt	ok
0101	20	0.13	4.2	0.018	0.13	0.37	0.55	6	68	2.1	681	1 564	0.31	2.20	1.51	31	16	stahl
0102	16	0.25	3.9	0.021	0.32	0.22	0.75	11	98	2.4	1 358	12 788	0.94	10.62	1.79	2	21	Bn
0103	20	0.18	4.3	0.027	0.33	0.24	0.47	11	38	4.0	440	738	0.19	2.66	1.25	388	21	fluenz
0104	16	0.26	4.7	0.004	0.26	0.23	0.27	11	43	2.8	1 389	6 056	0.70	2.22	1.99	0	21	stahl
0105	24	0.19	4.9	0.007	0.22	0.28	0.37	14	39	3.6	1 012	5 267	0.57	1.18	1.41	85	28	stahl
0106	24	0.27	4.8	0.007	0.42	0.29	0.30	6	94	1.5	1 568	11 316	0.94	0.92	2.02	0	18	stahl
0107	16	0.15	4.1	0.028	0.40	0.39	0.69	10	124	2.2	469	1 255	0.22	5.79	1.39	97	25	Bn
0108	24	0.17	3.9	0.022	0.44	0.17	0.44	7	28	2.2	965	5 738	0.49	2.90	1.06	2 590	18	fluenz
0109	24	0.22	4.6	0.022	0.25	0.38	0.41	9	49	3.7	1 323	22 747	0.92	1.47	1.67	6	22	dehn
0110	16	0.23	4.8	0.015	0.48	0.39	0.40	12	27	3.1	1 601	12 212	0.73	2.82	1.61	12	27	stahl
0111	16	0.30	3.8	0.026	0.46	0.10	0.65	11	86	2.0	1 627	10 386	0.90	8.08	2.12	0	20	stahl
0112	16	0.15	4.4	0.027	0.25	0.21	0.34	11	54	1.3	902	2 045	0.35	3.81	1.56	19	18	stahl
0113	16	0.14	3.8	0.016	0.55	0.26	0.47	14	47	3.6	535	1 625	0.22	9.10	1.22	563	24	Bn
0114	20	0.22	5.0	0.026	0.24	0.40	0.30	7	90	3.2	1 501	14 380	0.78	2.18	1.60	13	19	stahl
0115	24	0.21	3.9	0.014	0.57	0.39	0.18	6	41	3.6	698	2 102	0.34	1.62	1.20	633	15	fluenz
0116	24	0.14	5.0	0.011	0.53	0.30	0.69	7	33	3.8	1 129	7 726	0.63	1.60	0.98	6 219	26	fluenz
0117	16	0.30	3.6	0.003	0.42	0.40	0.37	6	114	1.2	2 648	18 309	2.32	12.15	2.16	0	14	dehn
0118	24	0.14	3.9	0.018	0.45	0.14	0.71	9	23	1.4	892	2 063	0.40	2.53	1.11	1 652	23	fluenz
0119	16	0.30	3.8	0.003	0.53	0.23	0.57	6	108	1.7	2 712	24 159	2.60	10.80	2.04	0	14	dehn
0120	20	0.24	3.7	0.014	0.33	0.29	0.22	12	81	1.8	976	1 870	0.38	3.98	1.77	2	15	stahl
0121*	24	0.13	4.3	0.019	0.47	0.31	0.54	14	20	1.5	286	749	0.11	2.15	0.96	7 686	36	fluenz
0122	24	0.12	4.5	0.005	0.45	0.36	0.55	12	74	1.8	278	694	0.13	1.10	1.13	1 377	30	fluenz
0123	16	0.20	4.0	0.025	0.11	0.13	0.44	12	92	3.6	757	1 315	0.34	6.51	1.68	5	15	Bn
0124	20	0.26	4.2	0.026	0.11	0.09	0.41	10	127	2.1	857	5 985	0.40	1.94	2.13	0	17	stahl
0125	20	0.26	4.6	0.007	0.40	0.25	0.64	11	144	2.3	855	5 765	0.50	2.03	1.93	0	25	stahl
0126	20	0.15	4.2	0.021	0.39	0.36	0.52	13	49	3.0	332	720	0.15	2.37	1.22	539	30	fluenz
0127	16	0.27	4.3	0.005	0.58	0.24	0.43	7	55	2.9	1 037	8 941	0.49	3.86	1.79	2	19	stahl
0128	24	0.15	4.0	0.013	0.51	0.27	0.59	11	72	3.2	611	2 438	0.32	1.99	1.07	2 361	30	fluenz
0129	24	0.24	4.4	0.018	0.30	0.28	0.63	12	69	3.4	1 411	6 464	0.82	1.42	1.80	2	26	stahl
0130	16	0.23	4.1	0.005	0.43	0.24	0.70	14	52	3.1	782	2 286	0.35	5.71	1.51	29	27	Bn
0131	20	0.20	4.7	0.008	0.39	0.17	0.29	14	151	2.0	1 598	20 042	0.91	1.44	1.43	64	25	stahl
0132	16	0.25	3.6	0.002	0.51	0.37	0.58	11	64	3.4	640	3 689	0.34	11.59	1.58	15	21	Bn
0133	20	0.26	4.8	0.023	0.60	0.09	0.80	7	24	1.8	2 313	362 657	1.31	2.80	1.69	5	23	stahl
0134	24	0.13	4.6	0.021	0.57	0.33	0.42	14	106	1.3	1 125	2 886	0.40	1.99	0.88	15 555	32	fluenz
0135	16	0.21	4.6	0.008	0.38	0.30	0.51	9	107	1.6	1 862	18 564	0.94	3.40	1.82	1	21	stahl
0136*	20	0.23	4.5	0.012	0.16	0.30	0.57	13	21	3.1	577	2 238	0.30	1.62	1.84	1	26	stahl
0137	24	0.16	3.9	0.001	0.18	0.35	0.28	6	39	1.6	612	2 562	0.29	0.94	1.57	17	16	stahl
0138	24	0.24	5.0	0.002	0.32	0.33	0.48	12	180	1.6	780	3 442	0.45	0.66	1.81	1	28	stahl
0139	20	0.29	4.8	0.013	0.52	0.33	0.48	11	36	2.4	2 862	30 000	1.82	2.82	2.02	0	26	dehn
0140	20	0.26	4.2	0.005	0.47	0.29	0.21	13	66	3.0	656	4 169	0.34	1.46	1.76	2	22	stahl
0141	16	0.18	4.0	0.018	0.40	0.34	0.16	13	26	3.4	884	2 092	0.32	6.73	1.41	81	20	Bn
0142	16	0.18	3.9	0.004	0.33	0.32	0.77	14	30	1.6	830	4 887	0.38	7.34	1.52	27	29	Bn
0143	20	0.17	4.8	0.006	0.43	0.28	0.36	10	115	2.3	761	1 973	0.28	1.25	1.34	172	22	stahl
0144	24	0.28	3.6	0.011	0.30	0.31	0.33	11	79	2.8	1 778	9 091	1.19	4.80	2.13	0	18	dehn
0145*	16	0.22	4.3	0.018	0.53	0.19	0.66	14	77	2.5	426	2 490	0.20	5.37	1.37	126	36	Bn
0146	24	0.23	4.2	0.005	0.54	0.33	0.60	12	181	3.2	962	7 187	0.57	1.14	1.45	56	33	stahl
0147	24	0.19	5.0	0.019	0.54	0.24	0.23	8	45	1.7	465	1 292	0.17	2.02	1.10	1 822	20	fluenz
0148	24	0.23	5.0	0.004	0.45	0.11	0.60	13	57	1.5	1 141	3 585	0.53	0.74	1.61	11	34	stahl
0149*	20	0.22	3.6	0.016	0.30	0.28	0.16	14	26	1.1	636	1 423	0.25	5.02	1.56	19	18	Bn
0150	20	0.28	3.8	0.000	0.37	0.10	0.68	7	134	2.2	2 129	31 817	1.41	6.81	2.07	0	15	dehn
0151	20	0.17	4.7	0.015	0.27	0.33	0.70	8	153	3.7	1 014	11 644	0.66	2.03	1.47	44	26	dehn
0152	24	0.25	4.6	0.010	0.58	0.27	0.56	14	33	3.3	1 024	4 796	0.68	0.99	1.66	7	40	dehn
0153	24	0.28	4.5	0.019	0.59	0.26	0.34	11	51	1.0	1 706	16 338	1.01	2.63	1.94	0	25	stahl
0154	20	0.29	4.0	0.019	0.34	0.12	0.31	10	61	2.3	2 565	21 004	1.33	5.82	2.22	0	14	stahl
0155	20	0.29	4.1	0.003	0.41	0.19	0.26	6	123	1.9	1 607	31 742	1.00	2.28	2.20	0	13	dehn
0156	16	0.26	4.0	0.014	0.56	0.21	0.77	11	105	2.9	623	3 243	0.31	6.25	1.66	7	26	Bn
0157	20	0.22	4.1	0.020	0.49	0.25	0.58	6	158	3.1	492	1 658	0.22	2.37	1.35	142	18	✓
0158	20	0.26	4.6	0.028	0.16	0.25	0.54	6	56	1.4	2 486	14 984	1.63	2.72	2.11	0	15	dehn
0159	24	0.14	4.3	0.026	0.11	0.15	0.33	9	134	3.4	1 922	4 529	0.73	2.62	1.29	279	13	stahl
0160	16	0.13	4.4	0.009	0.35	0.27	0.78	6	88	3.5	941	5 488	0.51	3.31	1.57	16	22	stahl
0161	24	0.27	4.0	0.007	0.48	0.18	0.75	14	29	1.1	1 102	19 494	0.91	1.66	1.94	0	32	dehn
0162	20	0.20	4.7	0.029	0.42	0.12	0.49	13	71	2.7	2 755	12 320	1.15	2.52	1.26	382	26	stahl
0163	24	0.16	4.7	0.018	0.47	0.21	0.76	9	45	1.4	1 347	6 590	0.64	1.89	1.06	2 701	29	fluenz
0164	20	0.26	4.8	0.022	0.32	0.18	0.39	7	43	2.3	1 810	7 235	0.84	2.14	1.99	0	17	stahl
0165	20	0.17	3.8	0.024	0.55	0.13	0.18	8	125	1.0	659	1 698	0.25	4.01	1.05	3 114	15	fluenz
0166	20	0.20	4.9	0.020	0.30	0.36	0.21	6	32	2.0	1 620	7 156	0.70	2.26	1.40	88	15	stahl
0167	24	0.23	4.5	0.004	0.37	0.26	0.80	11	30	1.7	1 064	4 862	0.65	0.89	1.63	9	34	dehn
0168	16	0.21	4.5	0.021	0.28	0.09	0.56	12	31	2.4	3 328	87 021	1.63	4.40	1.79	2	19	stahl
0169*	24	0.20	4.9	0.023	0.27	0.34	0.67	13	183	3.3	308	717	0.15	2.13	1.39	103	28	✓
0170	16	0.12	4.1	0.003	0.39	0.12	0.39	9	21	2.1	1 264	4 001	0.51	5.58	1.53	25	17	stahl
0171	24	0.20	4.9	0.001	0.15	0.15	0.17	10	102	1.2	829	1 952	0.33	0.56	1.54	22	17	stahl
0172	16	0.22	4.5	0.029	0.20	0.10	0.68	6	67	3.1	1 522	4 140	0.70	3.41	1.64	8	16	stahl
0173	24	0.22	4.2	0.028	0.57	0.20	0.41	13	28	2.3	533	1 827	0.24	1.85	1.28	298	27	stahl
0174	16	0.21	3.7	0.015	0.21	0.37	0.36	10	20	1.3	938	9 438	0.41	9.84	1.55	21	18	Bn
0175	16	0.27	4.0	0.011	0.17	0.16	0.67	10	53	2.5	976	8 082	0.56	6.41	2.15	0	17	Bn
0176*	20	0.19	3.6	0.016	0.42	0.34	0.65	14	165	1.1	434	2 674	0.22	4.86	1.20	689	32	fluenz
0177	24	0.24	4.4	0.019	0.31	0.18	0.47	10	129	2.6	867	4 068	0.45	1.38	1.78	2	19	stahl
0178	20	0.14	4.3	0.022	0.13	0.22	0.79	8	36	1.5	827	2 080	0.40	2.44	1.47	43	22	stahl
0179	16	0.28	4.2	0.008	0.54	0.11	0.33	7	28	1.8	1 612	12 248	0.71	4.34	1.88	1	16	stahl
0180	16	0.27	4.8	0.005	0.31	0.32	0.17	13	35	1.6	1 890	10 210	0.94	1.74	1.99	0	22	stahl
0181	24	0.15	4.9	0.025	0.21	0.21	0.42	9	189	3.9	782	2 109	0.32	2.14	1.23	508	18	stahl
0182	16	0.24	4.1	0.003	0.52	0.17	0.20	14	49	3.3	2 258	89 455	1.45	10.61	1.48	42	18	dehn
0183	20	0.20	4.5	0.024	0.13	0.39	0.35	6	83	2.6	1 555	11 544	0.85	2.36	1.57	16	15	stahl
0184	24	0.13	4.2	0.014	0.28	0.30	0.55	9	122	1.8	361	799	0.16	1.87	1.18	807	23	fluenz
0185	16	0.29	4.0	0.013	0.25	0.37	0.46	12	194	2.1	1 639	19 371	1.01	7.79	2.23	0	17	dehn
0186	20	0.14	4.4	0.029	0.53	0.12	0.61	8	34	2.7	512	1 244	0.20	2.76	1.04	3 147	24	fluenz
0187	16	0.16	4.9	0.005	0.22	0.36	0.66	13	58	2.3	1 094	4 736	0.51	1.73	1.77	2	34	stahl
0188	20	0.15	4.2	0.009	0.55	0.17	0.32	12	45	1.4	492	884	0.20	1.74	1.17	905	25	fluenz

No.	coils	dist. –	length –	force –	w.i. m	w.s. m	t m	n_c	E_c GPa	R_b m	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clear. m	heat W/m³	steel kt	ok
0189	16	0.30	4.1	0.016	0.54	0.32	0.76	7	99	3.9	1 510	17 307	1.21	9.07	2.08	0	20	dehn
0190	24	0.18	3.8	0.011	0.38	0.34	0.61	14	22	2.7	857	4 610	0.44	2.44	1.22	525	31	stahl
0191	24	0.16	4.4	0.013	0.20	0.35	0.37	11	101	1.3	1 494	9 072	0.86	1.91	1.34	166	19	stahl
0192	16	0.25	3.7	0.001	0.23	0.16	0.70	9	40	2.1	1 836	62 386	1.40	12.61	1.86	1	15	dehn
0193	16	0.13	3.8	0.008	0.36	0.20	0.28	11	76	3.4	585	1 284	0.25	8.57	1.45	57	16	Bn
0194	20	0.21	4.3	0.009	0.29	0.28	0.19	12	24	1.9	1 097	2 077	0.39	1.63	1.49	36	19	stahl
0195	24	0.12	3.8	0.023	0.34	0.23	0.62	10	80	3.5	359	932	0.15	2.36	1.08	2 264	22	fluenz
0196	16	0.15	4.0	0.030	0.15	0.24	0.61	8	200	3.9	672	1 529	0.29	6.33	1.63	9	15	Bn
0197	16	0.28	4.3	0.006	0.35	0.37	0.38	10	26	3.3	873	10 642	0.50	4.18	2.09	0	17	stahl
0198	20	0.18	5.0	0.028	0.25	0.09	0.58	9	35	2.5	1 930	18 667	0.93	2.45	1.37	119	18	stahl
0199	24	0.23	3.9	0.017	0.46	0.31	0.24	7	22	3.4	855	2 060	0.37	1.48	1.50	34	16	stahl

B Catalogue of all geometrically representable designs (list 2)

Geometrically representable means: the cases do not penetrate each other (check of the chain), and the PicoGK model confirms this. Each card shows the PicoGK model (80 mm voxels), top right the smallest reserve (negative = at least one limit violated) and below it the violated limits.

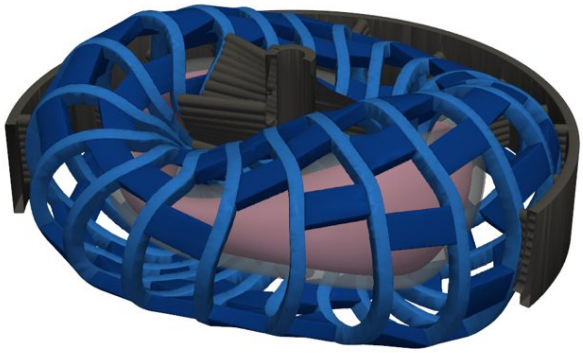
0000 −68 %



coils	24	σ_{99}	840 MPa
conn./pair	10	max.	3 425 MPa
E_V	70 GPa	strain	0.42 %
distance	0.18	$B \cdot n$	1.98 ‰
steel	23 kt	clear.	1.44 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

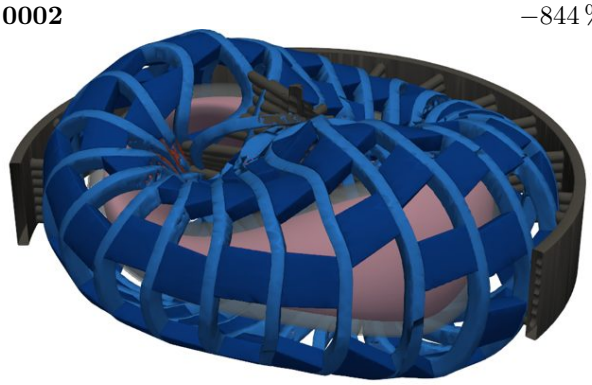
0001 −191 %



coils	20	σ_{99}	1 456 MPa
conn./pair	7	max.	3 851 MPa
E_V	135 GPa	strain	0.60 %
distance	0.24	$B \cdot n$	1.53 ‰
steel	12 kt	clear.	2.00 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

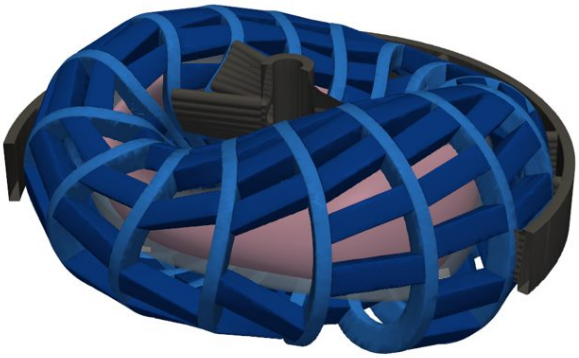
0002 −844 %



coils	24	σ_{99}	4 721 MPa
conn./pair	8	max.	13 357 MPa
E_V	25 GPa	strain	2.01 %
distance	0.30	$B \cdot n$	3.67 ‰
steel	13 kt	clear.	2.39 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

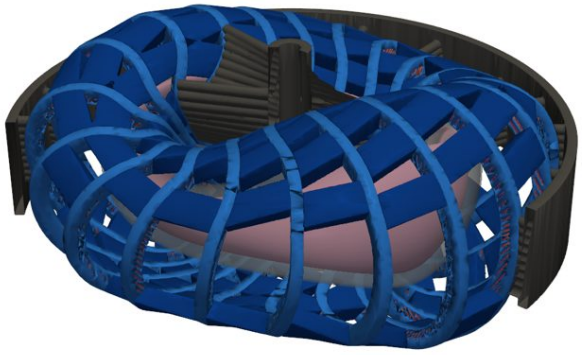
0003 −216 %



coils	16	σ_{99}	617 MPa
conn./pair	12	max.	1 361 MPa
E_V	142 GPa	strain	0.26 %
distance	0.20	$B \cdot n$	9.48 ‰
steel	20 kt	clear.	1.40 m

violated: steel σ99, field error B·n. PicoGK: no overlap.

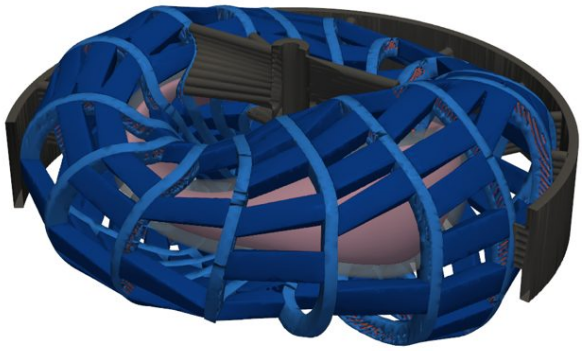
0004 −12 %



coils	20	σ_{99}	558 MPa
conn./pair	10	max.	1 085 MPa
E_V	148 GPa	strain	0.26 %
distance	0.19	$B \cdot n$	1.81 ‰
steel	19 kt	clear.	1.54 m

violated: steel σ99. PicoGK: no overlap.

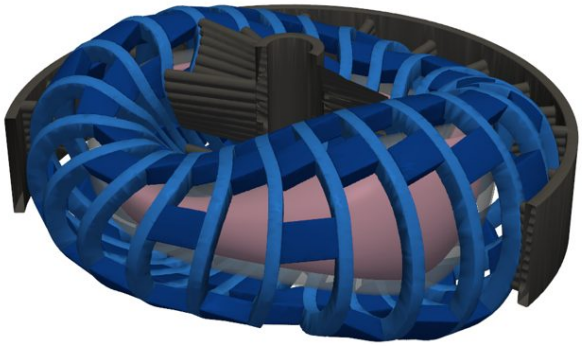
0005 −69 %



coils	16	σ_{99}	844 MPa
conn./pair	13	max.	1 847 MPa
E_V	114 GPa	strain	0.35 %
distance	0.15	$B \cdot n$	2.40 ‰
steel	23 kt	clear.	1.26 m

violated: steel σ99, winding-pack strain, neutron fluence. PicoGK: no overlap.

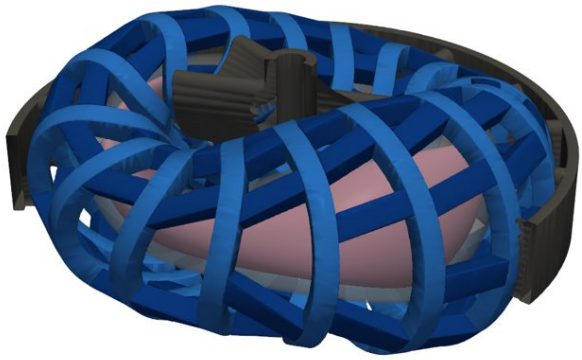
0006 −1 480 %



coils	24	σ_{99}	536 MPa
conn./pair	6	max.	1 222 MPa
E_V	78 GPa	strain	0.22 %
distance	0.12	$B \cdot n$	1.77 ‰
steel	14 kt	clear.	0.99 m

violated: steel σ99, package space, nuclear heating, neutron fluence. PicoGK: no overlap.

0007 −231 %



coils	16	σ_{99}	870 MPa
conn./pair	8	max.	3 574 MPa
E_V	88 GPa	strain	0.42 %
distance	0.16	$B \cdot n$	9.92 ‰
steel	17 kt	clear.	1.60 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

0008 27 %

coils	24	σ_{99}	305 MPa
conn./pair	11	max.	1 695 MPa
E_V	142 GPa	strain	0.17 %
distance	0.21	$B \cdot n$	2.20 ‰
steel	26 kt	clear.	1.48 m

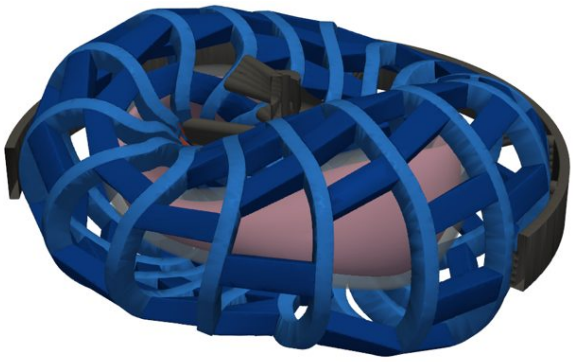
admissible. PicoGK: no overlap.

0009 −132 %

coils	24	σ_{99}	1 159 MPa
conn./pair	13	max.	9 216 MPa
E_V	145 GPa	strain	0.62 %
distance	0.23	$B \cdot n$	1.50 ‰
steel	24 kt	clear.	1.55 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

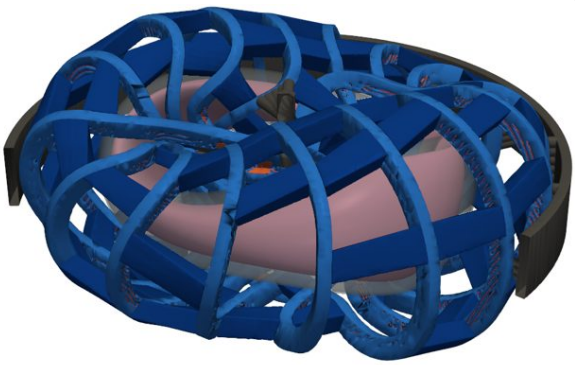
0010 −176 %



coils	16	σ_{99}	1 381 MPa
conn./pair	8	max.	8 739 MPa
E_V	149 GPa	strain	0.63 %
distance	0.27	$B \cdot n$	2.80 ‰
steel	21 kt	clear.	1.93 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

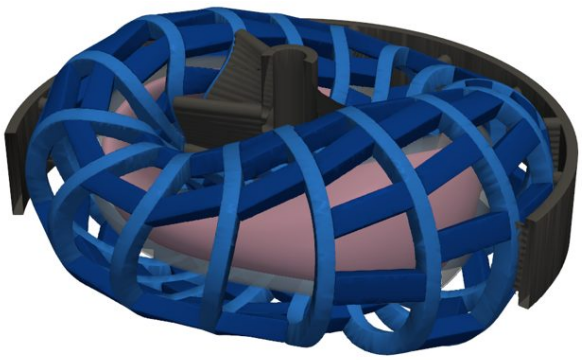
0011 −519 %



coils	16	σ_{99}	3 096 MPa
conn./pair	7	max.	19 166 MPa
E_V	42 GPa	strain	1.50 %
distance	0.18	$B \cdot n$	3.25 ‰
steel	14 kt	clear.	1.80 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

0012 −139 %



coils	16	σ_{99}	999 MPa
conn./pair	9	max.	2 952 MPa
E_V	50 GPa	strain	0.40 %
distance	0.21	$B \cdot n$	7.17 ‰
steel	13 kt	clear.	1.68 m

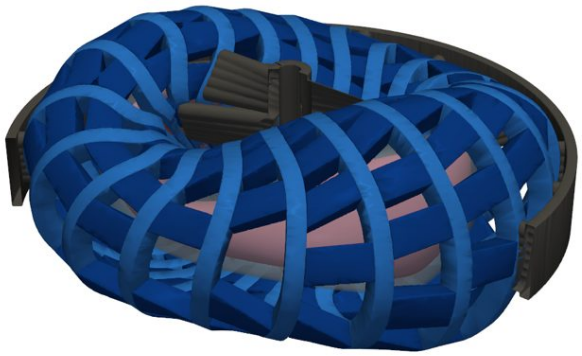
violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

0013 −55 %

coils	16	σ_{99}	773 MPa
conn./pair	14	max.	1 718 MPa
E_V	103 GPa	strain	0.34 %
distance	0.13	$B \cdot n$	4.32 ‰
steel	25 kt	clear.	1.63 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

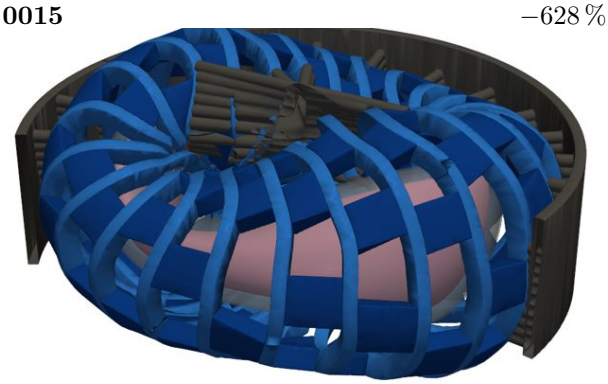
0014 25 %



coils	20	σ_{99}	377 MPa
conn./pair	11	max.	1 419 MPa
E_V	196 GPa	strain	0.21 %
distance	0.23	$B \cdot n$	2.21 ‰
steel	28 kt	clear.	1.65 m

admissible. PicoGK: no overlap.

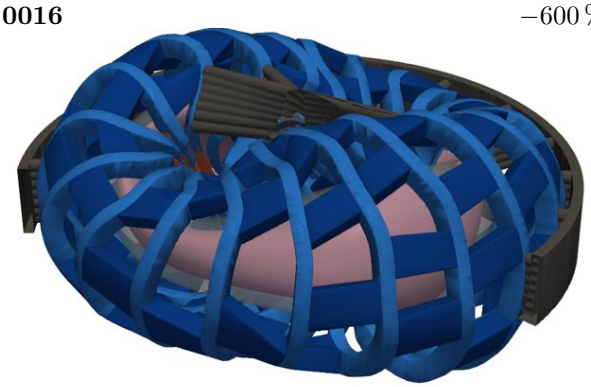
0015 −628 %



coils	24	σ_{99}	3 586 MPa
conn./pair	7	max.	30 683 MPa
E_V	168 GPa	strain	2.18 %
distance	0.29	$B \cdot n$	1.12 ‰
steel	18 kt	clear.	2.48 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

0016 −600 %



coils	20	σ_{99}	3 499 MPa
conn./pair	8	max.	15 129 MPa
E_V	172 GPa	strain	1.95 %
distance	0.29	$B \cdot n$	2.24 ‰
steel	19 kt	clear.	2.42 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

0017 20 %

coils	24	σ_{99}	389 MPa
conn./pair	7	max.	1 721 MPa
E_V	29 GPa	strain	0.20 %
distance	0.22	$B \cdot n$	1.66 ‰
steel	23 kt	clear.	1.30 m

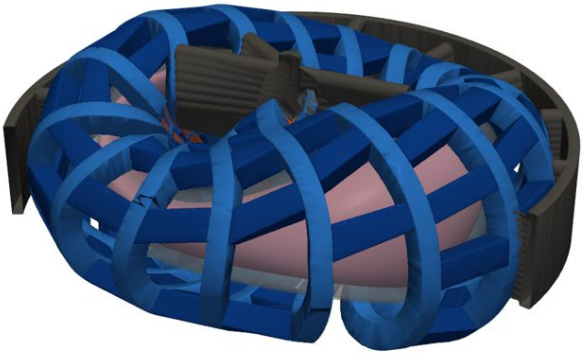
admissible. PicoGK: no overlap.

0018* −99 %

coils	20	σ_{99}	996 MPa
conn./pair	14	max.	4 566 MPa
E_V	26 GPa	strain	0.44 %
distance	0.25	$B \cdot n$	1.43 ‰
steel	24 kt	clear.	2.08 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

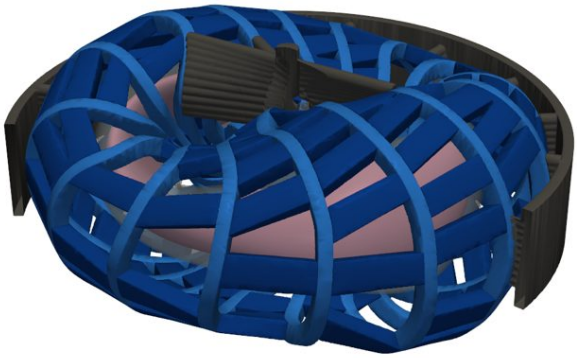
0019 −338 %



coils	16	σ_{99}	1 571 MPa
conn./pair	9	max.	14 989 MPa
E_V	24 GPa	strain	0.96 %
distance	0.25	$B \cdot n$	13.15 ‰
steel	17 kt	clear.	1.93 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

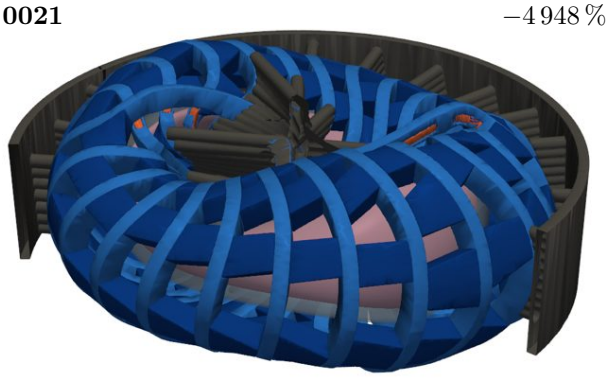
0020 −173 %



coils	16	σ_{99}	1 363 MPa
conn./pair	14	max.	6 952 MPa
E_V	137 GPa	strain	0.55 %
distance	0.24	$B \cdot n$	3.21 ‰
steel	17 kt	clear.	1.78 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

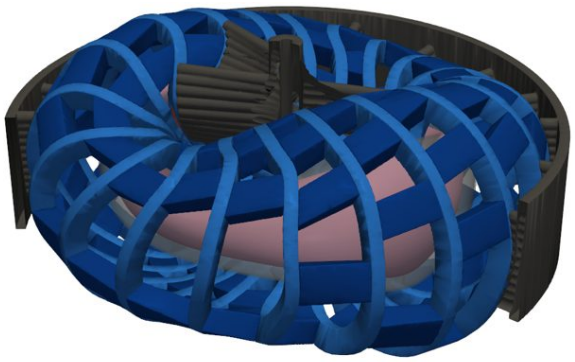
0021 −4 948 %



coils	24	σ_{99}	558 MPa
conn./pair	8	max.	4 173 MPa
E_V	138 GPa	strain	0.29 %
distance	0.16	$B \cdot n$	2.26 ‰
steel	26 kt	clear.	0.87 m

violated: steel σ99, package space, nuclear heating, neutron fluence. PicoGK: overlap!

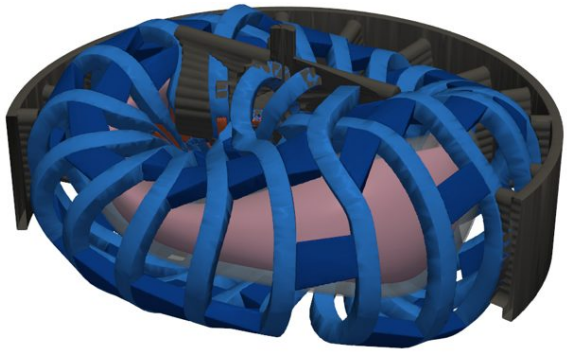
0022 −20 %



coils	20	σ_{99}	600 MPa
conn./pair	8	max.	2 183 MPa
E_V	37 GPa	strain	0.26 %
distance	0.23	$B \cdot n$	0.70 ‰
steel	24 kt	clear.	1.57 m

violated: steel σ99. PicoGK: no overlap.

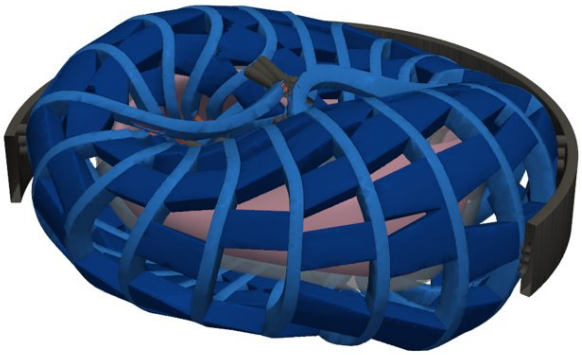
0023 −276 %



coils	20	σ_{99}	1 864 MPa
conn./pair	6	max.	10 858 MPa
E_V	156 GPa	strain	1.13 %
distance	0.28	$B \cdot n$	2.89 ‰
steel	15 kt	clear.	2.06 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

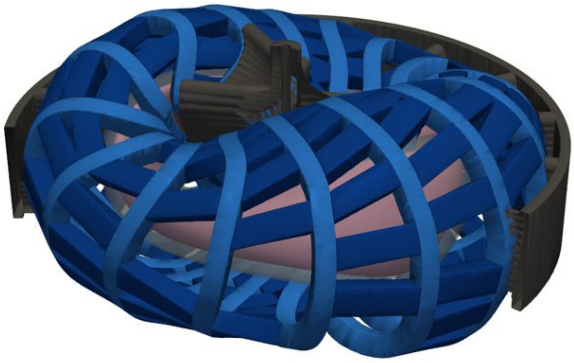
0024 −257 %



coils	20	σ_{99}	1 787 MPa
conn./pair	11	max.	7 614 MPa
E_V	24 GPa	strain	0.85 %
distance	0.26	$B \cdot n$	1.55 ‰
steel	24 kt	clear.	1.82 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

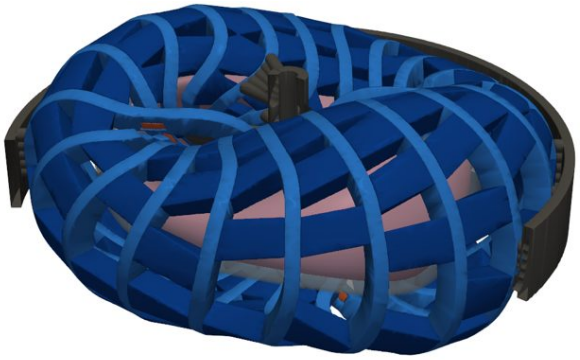
0025 −102 %



coils	16	σ_{99}	1 010 MPa
conn./pair	14	max.	9 842 MPa
E_V	41 GPa	strain	0.50 %
distance	0.18	$B \cdot n$	1.94 ‰
steel	26 kt	clear.	1.88 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

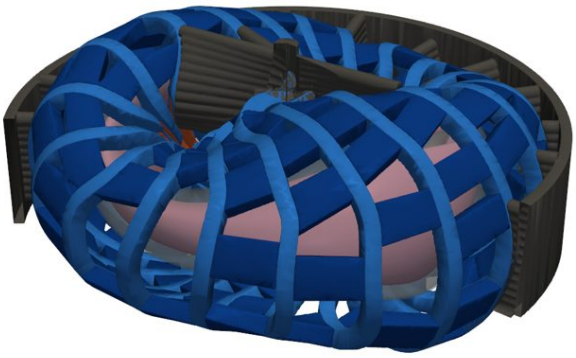
0026 −62 %



coils	20	σ_{99}	809 MPa
conn./pair	11	max.	2 253 MPa
E_V	79 GPa	strain	0.38 %
distance	0.19	$B \cdot n$	1.43 ‰
steel	29 kt	clear.	1.41 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

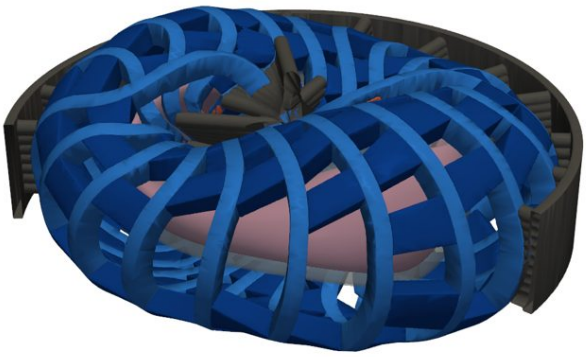
0027 −316 %



coils	20	σ_{99}	2 080 MPa
conn./pair	10	max.	10 143 MPa
E_V	171 GPa	strain	1.02 %
distance	0.28	$B \cdot n$	2.25 ‰
steel	16 kt	clear.	2.18 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

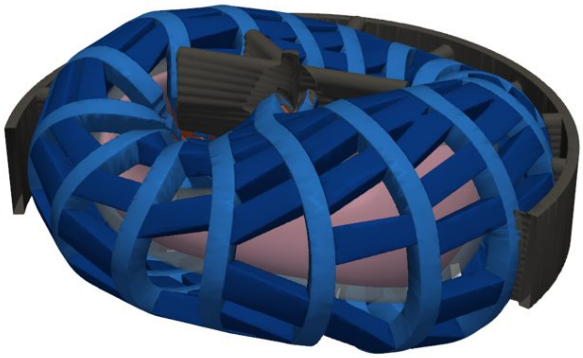
0028 −78 %



coils	20	σ_{99}	888 MPa
conn./pair	9	max.	3 682 MPa
E_V	22 GPa	strain	0.40 %
distance	0.17	$B \cdot n$	1.45 ‰
steel	30 kt	clear.	1.33 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

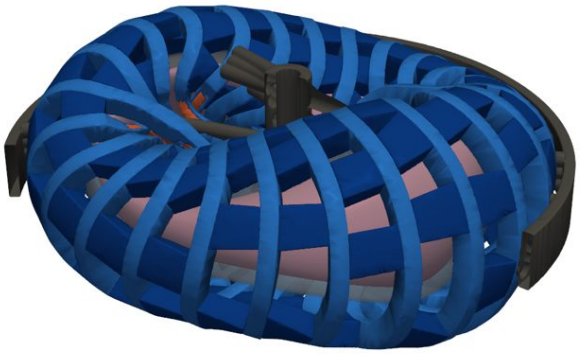
0029 −361 %



coils	16	σ_{99}	1 975 MPa
conn./pair	13	max.	34 584 MPa
E_V	27 GPa	strain	1.38 %
distance	0.29	$B \cdot n$	10.34 ‰
steel	22 kt	clear.	2.35 m

violated: steel σ99, winding-pack strain, field error B·n. PicoGK: no overlap.

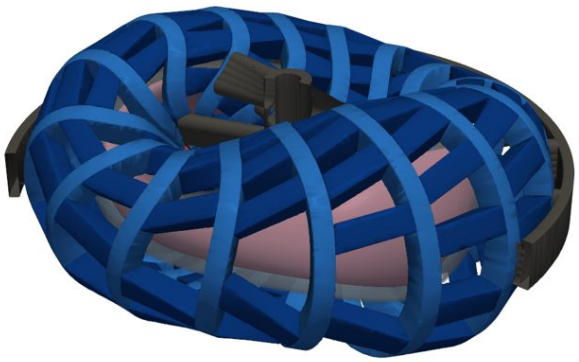
0031 −222 %



coils	24	σ_{99}	1 514 MPa
conn./pair	8	max.	8 426 MPa
E_V	190 GPa	strain	0.97 %
distance	0.24	$B \cdot n$	1.99 ‰
steel	21 kt	clear.	1.84 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

0032 −123 %



coils	16	σ_{99}	777 MPa
conn./pair	12	max.	3 008 MPa
E_V	38 GPa	strain	0.28 %
distance	0.17	$B \cdot n$	6.70 ‰
steel	25 kt	clear.	1.48 m

violated: steel σ99, field error B·n. PicoGK: no overlap.

0033 −125 %

coils	24	σ_{99}	1 019 MPa
conn./pair	8	max.	4 409 MPa
E_V	118 GPa	strain	0.68 %
distance	0.25	$B \cdot n$	1.78 ‰
steel	25 kt	clear.	1.56 m

violated: steel σ99, winding-pack strain. PicoGK: no overlap.

0034 −215 %

coils	16	σ_{99}	1 574 MPa
conn./pair	6	max.	4 775 MPa
E_V	21 GPa	strain	0.56 %
distance	0.26	$B \cdot n$	3.08 ‰
steel	17 kt	clear.	1.96 m

violated: steel σ99, winding-pack strain, field error B·n.

0035 −278 %

coils	24	σ_{99}	1 892 MPa
conn./pair	12	max.	5 956 MPa
E_V	50 GPa	strain	0.83 %
distance	0.18	$B \cdot n$	1.23 ‰
steel	30 kt	clear.	1.31 m

violated: steel σ99, winding-pack strain.

0036 −37 %

coils	20	σ_{99}	683 MPa
conn./pair	7	max.	7 457 MPa
E_V	119 GPa	strain	0.38 %
distance	0.25	$B \cdot n$	3.45 ‰
steel	15 kt	clear.	1.74 m

violated: steel σ99, winding-pack strain, field error B·n.

0037 −260 %

coils	24	σ_{99}	1 800 MPa
conn./pair	6	max.	4 950 MPa
E_V	53 GPa	strain	0.66 %
distance	0.13	$B \cdot n$	1.33 ‰
steel	12 kt	clear.	1.43 m

violated: steel σ99, winding-pack strain.

0038 −204 %

coils	20	σ_{99}	525 MPa
conn./pair	13	max.	2 561 MPa
E_V	60 GPa	strain	0.23 %
distance	0.17	$B \cdot n$	3.27 ‰
steel	32 kt	clear.	1.15 m

violated: steel σ99, neutron fluence, field error B·n.

0039 −210 %

coils	16	σ_{99}	1 548 MPa
conn./pair	7	max.	6 664 MPa
E_V	31 GPa	strain	0.70 %
distance	0.28	$B \cdot n$	4.38 ‰
steel	14 kt	clear.	2.05 m

violated: steel σ99, winding-pack strain, field error B·n.

0040 −240 %

coils	24	σ_{99}	1 262 MPa
conn./pair	12	max.	9 182 MPa
E_V	91 GPa	strain	1.02 %
distance	0.29	$B \cdot n$	3.84 ‰
steel	23 kt	clear.	2.41 m

violated: steel σ99, winding-pack strain, field error B·n.

0041 −69 %

coils	16	σ_{99}	843 MPa
conn./pair	9	max.	1 629 MPa
E_V	73 GPa	strain	0.36 %
distance	0.14	$B \cdot n$	4.96 ‰
steel	13 kt	clear.	1.60 m

violated: steel σ99, winding-pack strain, field error B·n.

0042 −26 %

coils	20	σ_{99}	632 MPa
conn./pair	12	max.	3 962 MPa
E_V	170 GPa	strain	0.34 %
distance	0.14	$B \cdot n$	2.66 ‰
steel	21 kt	clear.	1.61 m

violated: steel σ99, winding-pack strain.

0043 −31 %

coils	24	σ_{99}	653 MPa
conn./pair	9	max.	1 208 MPa
E_V	40 GPa	strain	0.29 %
distance	0.17	$B \cdot n$	1.79 ‰
steel	16 kt	clear.	1.30 m

violated: steel σ99.

0044 −8 %

coils	20	σ_{99}	540 MPa
conn./pair	13	max.	1 583 MPa
E_V	110 GPa	strain	0.23 %
distance	0.16	$B \cdot n$	2.76 ‰
steel	26 kt	clear.	1.35 m

violated: steel σ99.

0045 −58 %

coils	20	σ_{99}	600 MPa
conn./pair	9	max.	3 060 MPa
E_V	175 GPa	strain	0.29 %
distance	0.15	$B \cdot n$	4.74 ‰
steel	22 kt	clear.	1.36 m

violated: steel σ99, field error B·n.

0046 −112 %

coils	16	σ_{99}	619 MPa
conn./pair	9	max.	2 502 MPa
E_V	46 GPa	strain	0.26 %
distance	0.16	$B \cdot n$	6.36 ‰
steel	22 kt	clear.	1.43 m

violated: steel σ99, field error B·n.

0047 −407 %

coils	24	σ_{99}	983 MPa
conn./pair	8	max.	4 736 MPa
E_V	34 GPa	strain	0.42 %
distance	0.15	$B \cdot n$	1.48 ‰
steel	25 kt	clear.	1.10 m

violated: steel σ99, winding-pack strain, package space, neutron fluence.

0048 −364 %

coils	16	σ_{99}	2 081 MPa
conn./pair	7	max.	19 844 MPa
E_V	160 GPa	strain	1.39 %
distance	0.29	$B \cdot n$	10.18 ‰
steel	13 kt	clear.	2.32 m

violated: steel σ99, winding-pack strain, field error B·n.

0049 −462 %

coils	16	σ_{99}	2 808 MPa
conn./pair	9	max.	9 212 MPa
E_V	84 GPa	strain	1.07 %
distance	0.21	$B \cdot n$	2.42 ‰
steel	18 kt	clear.	1.74 m

violated: steel σ99, winding-pack strain.

0051 −130 %

coils	20	σ_{99}	1 152 MPa
conn./pair	10	max.	2 169 MPa
E_V	37 GPa	strain	0.50 %
distance	0.17	$B \cdot n$	3.32 ‰
steel	15 kt	clear.	1.50 m

violated: steel σ99, winding-pack strain, field error B·n.

0052 −164 %

coils	24	σ_{99}	1 321 MPa
conn./pair	13	max.	4 323 MPa
E_V	55 GPa	strain	0.52 %
distance	0.19	$B \cdot n$	2.06 ‰
steel	20 kt	clear.	1.39 m

violated: steel σ99, winding-pack strain.

0053 −163 %

coils	20	σ_{99}	1 314 MPa
conn./pair	6	max.	9 639 MPa
E_V	57 GPa	strain	0.57 %
distance	0.25	$B \cdot n$	1.71 ‰
steel	11 kt	clear.	2.02 m

violated: steel σ99, winding-pack strain.

0054 −138 %

coils	16	σ_{99}	963 MPa
conn./pair	7	max.	2 479 MPa
E_V	62 GPa	strain	0.37 %
distance	0.16	$B \cdot n$	7.15 ‰
steel	12 kt	clear.	1.42 m

violated: steel σ99, winding-pack strain, field error B·n.

0055* −192 %

coils	20	σ_{99}	413 MPa
conn./pair	14	max.	942 MPa
E_V	38 GPa	strain	0.18 %
distance	0.13	$B \cdot n$	2.84 ‰
steel	32 kt	clear.	1.15 m

violated: neutron fluence.

0056 −423 %

coils	20	σ_{99}	2 615 MPa
conn./pair	8	max.	13 038 MPa
E_V	59 GPa	strain	1.21 %
distance	0.28	$B \cdot n$	2.25 ‰
steel	16 kt	clear.	2.06 m

violated: steel σ99, winding-pack strain.

0057 −111 %

coils	20	σ_{99}	1 056 MPa
conn./pair	6	max.	4 063 MPa
E_V	31 GPa	strain	0.39 %
distance	0.24	$B \cdot n$	2.18 ‰
steel	14 kt	clear.	1.60 m

violated: steel σ99, winding-pack strain.

0058 −124 %

coils	16	σ_{99}	689 MPa
conn./pair	10	max.	2 099 MPa
E_V	32 GPa	strain	0.29 %
distance	0.23	$B \cdot n$	6.71 ‰
steel	29 kt	clear.	1.60 m

violated: steel σ99, field error B·n.

0059 −132 %

coils	16	σ_{99}	1 161 MPa
conn./pair	11	max.	4 304 MPa
E_V	110 GPa	strain	0.57 %
distance	0.19	$B \cdot n$	2.27 ‰
steel	31 kt	clear.	1.48 m

violated: steel σ99, winding-pack strain.

0060* −41 %

coils	24	σ_{99}	704 MPa
conn./pair	13	max.	1 499 MPa
E_V	155 GPa	strain	0.29 %
distance	0.22	$B \cdot n$	1.61 ‰
steel	16 kt	clear.	1.66 m

violated: steel σ99.

0061 30 %

coils	24	σ_{99}	351 MPa
conn./pair	10	max.	725 MPa
E_V	93 GPa	strain	0.17 %
distance	0.20	$B \cdot n$	1.49 ‰
steel	19 kt	clear.	1.62 m

admissible.

0062 −427 %

coils	24	σ_{99}	2 627 MPa
conn./pair	9	max.	13 807 MPa
E_V	21 GPa	strain	1.58 %
distance	0.30	$B \cdot n$	3.27 ‰
steel	17 kt	clear.	2.40 m

violated: steel σ99, winding-pack strain, field error B·n.

0063* −24 %

coils	20	σ_{99}	595 MPa
conn./pair	14	max.	4 265 MPa
E_V	43 GPa	strain	0.37 %
distance	0.18	$B \cdot n$	0.75 ‰
steel	41 kt	clear.	1.53 m

violated: steel σ99, winding-pack strain.

0064 −90 %

coils	16	σ_{99}	683 MPa
conn./pair	12	max.	5 601 MPa
E_V	131 GPa	strain	0.35 %
distance	0.27	$B \cdot n$	5.70 ‰
steel	22 kt	clear.	1.85 m

violated: steel σ99, winding-pack strain, field error B·n.

0065 −32 %

coils	20	σ_{99}	660 MPa
conn./pair	9	max.	1 210 MPa
E_V	48 GPa	strain	0.31 %
distance	0.17	$B \cdot n$	1.97 ‰
steel	15 kt	clear.	1.61 m

violated: steel σ99, winding-pack strain.

0067 −100 %

coils	20	σ_{99}	999 MPa
conn./pair	11	max.	2 826 MPa
E_V	82 GPa	strain	0.47 %
distance	0.20	$B \cdot n$	3.08 ‰
steel	28 kt	clear.	1.53 m

violated: steel σ99, winding-pack strain, field error B·n.

0068 −102 %

coils	20	σ_{99}	1 012 MPa
conn./pair	10	max.	2 393 MPa
E_V	75 GPa	strain	0.42 %
distance	0.12	$B \cdot n$	1.57 ‰
steel	18 kt	clear.	1.53 m

violated: steel σ99, winding-pack strain.

0069 −225 %

coils	20	σ_{99}	1 624 MPa
conn./pair	7	max.	16 704 MPa
E_V	130 GPa	strain	0.84 %
distance	0.29	$B \cdot n$	6.85 ‰
steel	15 kt	clear.	2.16 m

violated: steel σ99, winding-pack strain, field error B·n.

0070 −134 %

coils	24	σ_{99}	1 172 MPa
conn./pair	14	max.	6 574 MPa
E_V	119 GPa	strain	0.56 %
distance	0.18	$B \cdot n$	0.83 ‰
steel	28 kt	clear.	1.51 m

violated: steel σ99, winding-pack strain.

0071 −121 %

coils	24	σ_{99}	1 075 MPa
conn./pair	12	max.	5 955 MPa
E_V	192 GPa	strain	0.66 %
distance	0.25	$B \cdot n$	1.55 ‰
steel	30 kt	clear.	1.61 m

violated: steel σ99, winding-pack strain.

0072 −119 %

coils	24	σ_{99}	1 093 MPa
conn./pair	9	max.	8 828 MPa
E_V	95 GPa	strain	0.56 %
distance	0.17	$B \cdot n$	2.35 ‰
steel	22 kt	clear.	1.33 m

violated: steel σ99, winding-pack strain.

0073 −363 %

coils	16	σ_{99}	1 030 MPa
conn./pair	12	max.	50 101 MPa
E_V	52 GPa	strain	0.71 %
distance	0.24	$B \cdot n$	13.89 ‰
steel	23 kt	clear.	1.54 m

violated: steel σ99, winding-pack strain, field error B·n.

0074 −243 %

coils	16	σ_{99}	1 715 MPa
conn./pair	6	max.	23 336 MPa
E_V	85 GPa	strain	0.65 %
distance	0.26	$B \cdot n$	2.71 ‰
steel	14 kt	clear.	1.74 m

violated: steel σ99, winding-pack strain.

0075 −54 %

coils	20	σ_{99}	477 MPa
conn./pair	10	max.	1 064 MPa
E_V	163 GPa	strain	0.20 %
distance	0.20	$B \cdot n$	4.62 ‰
steel	18 kt	clear.	1.25 m

violated: neutron fluence, field error B·n.

0076 −350 %

coils	20	σ_{99}	1 983 MPa
conn./pair	9	max.	10 425 MPa
E_V	33 GPa	strain	1.35 %
distance	0.27	$B \cdot n$	2.07 ‰
steel	23 kt	clear.	2.13 m

violated: steel σ99, winding-pack strain.

0077 −56 %

coils	24	σ_{99}	779 MPa
conn./pair	10	max.	1 288 MPa
E_V	47 GPa	strain	0.31 %
distance	0.21	$B \cdot n$	1.07 ‰
steel	14 kt	clear.	1.65 m

violated: steel σ99, winding-pack strain.

0078 −284 %

coils	24	σ_{99}	1 919 MPa
conn./pair	8	max.	14 222 MPa
E_V	36 GPa	strain	0.85 %
distance	0.24	$B \cdot n$	1.52 ‰
steel	18 kt	clear.	1.85 m

violated: steel σ99, winding-pack strain.

0079 −164 %

coils	20	σ_{99}	1 320 MPa
conn./pair	13	max.	8 604 MPa
E_V	30 GPa	strain	0.66 %
distance	0.27	$B \cdot n$	1.17 ‰
steel	23 kt	clear.	2.00 m

violated: steel σ99, winding-pack strain.

0080 −61 %

coils	16	σ_{99}	803 MPa
conn./pair	12	max.	1 759 MPa
E_V	65 GPa	strain	0.32 %
distance	0.13	$B \cdot n$	3.98 ‰
steel	21 kt	clear.	1.47 m

violated: steel σ99, winding-pack strain, field error B·n.

0081 −36 %

coils	24	σ_{99}	577 MPa
conn./pair	9	max.	5 361 MPa
E_V	97 GPa	strain	0.41 %
distance	0.27	$B \cdot n$	2.32 ‰
steel	19 kt	clear.	2.17 m

violated: steel σ99, winding-pack strain.

0082 −418 %

coils	20	σ_{99}	2 591 MPa
conn./pair	8	max.	34 831 MPa
E_V	23 GPa	strain	1.50 %
distance	0.16	$B \cdot n$	1.75 ‰
steel	17 kt	clear.	1.72 m

violated: steel σ99, winding-pack strain.

0083 −99 %

coils	16	σ_{99}	996 MPa
conn./pair	6	max.	5 784 MPa
E_V	27 GPa	strain	0.54 %
distance	0.26	$B \cdot n$	5.30 ‰
steel	14 kt	clear.	2.05 m

violated: steel σ99, winding-pack strain, field error B·n.

0084* −225 %

coils	16	σ_{99}	462 MPa
conn./pair	14	max.	1 018 MPa
E_V	63 GPa	strain	0.20 %
distance	0.19	$B \cdot n$	9.75 ‰
steel	20 kt	clear.	1.26 m

violated: neutron fluence, field error B·n.

0085 −31 %

coils	20	σ_{99}	608 MPa
conn./pair	10	max.	2 067 MPa
E_V	22 GPa	strain	0.28 %
distance	0.19	$B \cdot n$	2.00 ‰
steel	30 kt	clear.	1.23 m

violated: steel σ99, neutron fluence.

0086 −168 %

coils	16	σ_{99}	1 339 MPa
conn./pair	13	max.	5 294 MPa
E_V	67 GPa	strain	0.53 %
distance	0.20	$B \cdot n$	2.87 ‰
steel	21 kt	clear.	1.64 m

violated: steel σ99, winding-pack strain.

0087 −267 %

coils	20	σ_{99}	1 837 MPa
conn./pair	6	max.	11 946 MPa
E_V	30 GPa	strain	0.94 %
distance	0.21	$B \cdot n$	1.40 ‰
steel	17 kt	clear.	1.32 m

violated: steel σ99, winding-pack strain.

0088 −3 286 %

coils	24	σ_{99}	1 478 MPa
conn./pair	13	max.	5 859 MPa
E_V	86 GPa	strain	0.71 %
distance	0.14	$B \cdot n$	1.92 ‰
steel	33 kt	clear.	0.91 m

violated: steel σ99, winding-pack strain, package space, nuclear heating, neutron fluence.

0089 −546 %

coils	24	σ_{99}	645 MPa
conn./pair	12	max.	2 827 MPa
E_V	22 GPa	strain	0.30 %
distance	0.15	$B \cdot n$	2.16 ‰
steel	30 kt	clear.	1.07 m

violated: steel σ99, package space, nuclear heating, neutron fluence.

0090 −286 %

coils	20	σ_{99}	1 929 MPa
conn./pair	6	max.	10 870 MPa
E_V	161 GPa	strain	0.91 %
distance	0.19	$B \cdot n$	2.84 ‰
steel	17 kt	clear.	1.36 m

violated: steel σ99, winding-pack strain.

0091 −70 %

coils	16	σ_{99}	848 MPa
conn./pair	10	max.	2 510 MPa
E_V	42 GPa	strain	0.38 %
distance	0.22	$B \cdot n$	4.47 ‰
steel	23 kt	clear.	1.52 m

violated: steel σ99, winding-pack strain, field error B·n.

0092 −151 %

coils	20	σ_{99}	1 195 MPa
conn./pair	10	max.	14 134 MPa
E_V	44 GPa	strain	0.75 %
distance	0.23	$B \cdot n$	3.37 ‰
steel	23 kt	clear.	1.43 m

violated: steel σ99, winding-pack strain, field error B·n.

0093 27 %

coils	24	σ_{99}	283 MPa
conn./pair	8	max.	1 190 MPa
E_V	64 GPa	strain	0.16 %
distance	0.21	$B \cdot n$	1.45 ‰
steel	28 kt	clear.	1.37 m

admissible.

0095 −55 %

coils	16	σ_{99}	773 MPa
conn./pair	8	max.	2 138 MPa
E_V	186 GPa	strain	0.32 %
distance	0.19	$B \cdot n$	3.25 ‰
steel	22 kt	clear.	1.28 m

violated: steel σ99, winding-pack strain, field error B·n.

0096 −185 %

coils	16	σ_{99}	1 006 MPa
conn./pair	11	max.	3 209 MPa
E_V	25 GPa	strain	0.40 %
distance	0.25	$B \cdot n$	8.54 ‰
steel	15 kt	clear.	1.85 m

violated: steel σ99, winding-pack strain, field error B·n.

0097 −331 %

coils	24	σ_{99}	2 155 MPa
conn./pair	7	max.	28 026 MPa
E_V	99 GPa	strain	0.95 %
distance	0.14	$B \cdot n$	2.76 ‰
steel	14 kt	clear.	1.30 m

violated: steel σ99, winding-pack strain.

0098 −350 %

coils	20	σ_{99}	2 250 MPa
conn./pair	13	max.	13 304 MPa
E_V	25 GPa	strain	1.30 %
distance	0.29	$B \cdot n$	6.96 ‰
steel	20 kt	clear.	2.18 m

violated: steel σ99, winding-pack strain, field error B·n.

0099 −305 %

coils	20	σ_{99}	2 025 MPa
conn./pair	8	max.	7 095 MPa
E_V	62 GPa	strain	0.97 %
distance	0.28	$B \cdot n$	2.32 ‰
steel	20 kt	clear.	1.89 m

violated: steel σ99, winding-pack strain.

0100 −103 %

coils	24	σ_{99}	441 MPa
conn./pair	7	max.	788 MPa
E_V	59 GPa	strain	0.20 %
distance	0.13	$B \cdot n$	2.37 ‰
steel	19 kt	clear.	1.19 m

violated: neutron fluence.

0101 −36 %

coils	20	σ_{99}	681 MPa
conn./pair	6	max.	1 564 MPa
E_V	68 GPa	strain	0.31 %
distance	0.13	$B \cdot n$	2.20 ‰
steel	16 kt	clear.	1.51 m

violated: steel σ99, winding-pack strain.

0102 −254 %

coils	16	σ_{99}	1 358 MPa
conn./pair	11	max.	12 788 MPa
E_V	98 GPa	strain	0.94 %
distance	0.25	$B \cdot n$	10.62 ‰
steel	21 kt	clear.	1.79 m

violated: steel σ99, winding-pack strain, field error B·n.

0103 −8 %

coils	20	σ_{99}	440 MPa
conn./pair	11	max.	738 MPa
E_V	38 GPa	strain	0.19 %
distance	0.18	$B \cdot n$	2.66 ‰
steel	21 kt	clear.	1.25 m

violated: neutron fluence.

0104 −178 %

coils	16	σ_{99}	1 389 MPa
conn./pair	11	max.	6 056 MPa
E_V	43 GPa	strain	0.70 %
distance	0.26	$B \cdot n$	2.22 ‰
steel	21 kt	clear.	1.99 m

violated: steel σ99, winding-pack strain.

0105 −102 %

coils	24	σ_{99}	1 012 MPa
conn./pair	14	max.	5 267 MPa
E_V	39 GPa	strain	0.57 %
distance	0.19	$B \cdot n$	1.18 ‰
steel	28 kt	clear.	1.41 m

violated: steel σ99, winding-pack strain.

0106 −214 %

coils	24	σ_{99}	1 568 MPa
conn./pair	6	max.	11 316 MPa
E_V	94 GPa	strain	0.94 %
distance	0.27	$B \cdot n$	0.92 ‰
steel	18 kt	clear.	2.02 m

violated: steel σ99, winding-pack strain.

0107 −93 %

coils	16	σ_{99}	469 MPa
conn./pair	10	max.	1 255 MPa
E_V	124 GPa	strain	0.22 %
distance	0.15	$B \cdot n$	5.79 ‰
steel	25 kt	clear.	1.39 m

violated: field error B·n.

0110 −220 %

coils	16	σ_{99}	1 601 MPa
conn./pair	12	max.	12 212 MPa
E_V	27 GPa	strain	0.73 %
distance	0.23	$B \cdot n$	2.82 ‰
steel	27 kt	clear.	1.61 m

violated: steel σ99, winding-pack strain.

0111 −225 %

coils	16	σ_{99}	1 627 MPa
conn./pair	11	max.	10 386 MPa
E_V	86 GPa	strain	0.90 %
distance	0.30	$B \cdot n$	8.08 ‰
steel	20 kt	clear.	2.12 m

violated: steel σ99, winding-pack strain, field error B·n.

0112 −80 %

coils	16	σ_{99}	902 MPa
conn./pair	11	max.	2 045 MPa
E_V	54 GPa	strain	0.35 %
distance	0.15	$B \cdot n$	3.81 ‰
steel	18 kt	clear.	1.56 m

violated: steel σ99, winding-pack strain, field error B·n.

0113 −203 %

coils	16	σ_{99}	535 MPa
conn./pair	14	max.	1 625 MPa
E_V	47 GPa	strain	0.22 %
distance	0.14	$B \cdot n$	9.10 ‰
steel	24 kt	clear.	1.22 m

violated: steel σ99, neutron fluence, field error B·n.

0114 −200 %

coils	20	σ_{99}	1 501 MPa
conn./pair	7	max.	14 380 MPa
E_V	90 GPa	strain	0.78 %
distance	0.22	$B \cdot n$	2.18 ‰
steel	19 kt	clear.	1.60 m

violated: steel σ99, winding-pack strain.

0115 −76 %

coils	24	σ_{99}	698 MPa
conn./pair	6	max.	2 102 MPa
E_V	41 GPa	strain	0.34 %
distance	0.21	$B \cdot n$	1.62 ‰
steel	15 kt	clear.	1.20 m

violated: steel σ99, winding-pack strain, neutron fluence.

0116 −1 627 %

coils	24	σ_{99}	1 129 MPa
conn./pair	7	max.	7 726 MPa
E_V	33 GPa	strain	0.63 %
distance	0.14	$B \cdot n$	1.60 ‰
steel	26 kt	clear.	0.98 m

violated: steel σ99, winding-pack strain, package space, nuclear heating, neutron fluence.

0117 −675 %

coils	16	σ_{99}	2 648 MPa
conn./pair	6	max.	18 309 MPa
E_V	114 GPa	strain	2.32 %
distance	0.30	$B \cdot n$	12.15 ‰
steel	14 kt	clear.	2.16 m

violated: steel σ99, winding-pack strain, field error B·n.

0118 −359 %

coils	24	σ_{99}	892 MPa
conn./pair	9	max.	2 063 MPa
E_V	23 GPa	strain	0.40 %
distance	0.14	$B \cdot n$	2.53 ‰
steel	23 kt	clear.	1.11 m

violated: steel σ99, winding-pack strain, neutron fluence.

0119 −767 %

coils	16	σ_{99}	2 712 MPa
conn./pair	6	max.	24 159 MPa
E_V	108 GPa	strain	2.60 %
distance	0.30	$B \cdot n$	10.80 ‰
steel	14 kt	clear.	2.04 m

violated: steel σ99, winding-pack strain, field error B·n.

0121* −2 035 %

coils	24	σ_{99}	286 MPa
conn./pair	14	max.	749 MPa
E_V	20 GPa	strain	0.11 %
distance	0.13	$B \cdot n$	2.15 ‰
steel	36 kt	clear.	0.96 m

violated: package space, nuclear heating, neutron fluence.

0122 −282 %

coils	24	σ_{99}	278 MPa
conn./pair	12	max.	694 MPa
E_V	74 GPa	strain	0.13 %
distance	0.12	$B \cdot n$	1.10 ‰
steel	30 kt	clear.	1.13 m

violated: neutron fluence.

0123 −117 %

coils	16	σ_{99}	757 MPa
conn./pair	12	max.	1 315 MPa
E_V	92 GPa	strain	0.34 %
distance	0.20	$B \cdot n$	6.51 ‰
steel	15 kt	clear.	1.68 m

violated: steel σ99, winding-pack strain, field error B·n.

0124 −71 %

coils	20	σ_{99}	857 MPa
conn./pair	10	max.	5 985 MPa
E_V	127 GPa	strain	0.40 %
distance	0.26	$B \cdot n$	1.94 ‰
steel	17 kt	clear.	2.13 m

violated: steel σ99, winding-pack strain.

0125 −71 %

coils	20	σ_{99}	855 MPa
conn./pair	11	max.	5 765 MPa
E_V	144 GPa	strain	0.50 %
distance	0.26	$B \cdot n$	2.03 ‰
steel	25 kt	clear.	1.93 m

violated: steel σ99, winding-pack strain.

0126 −50 %

coils	20	σ_{99}	332 MPa
conn./pair	13	max.	720 MPa
E_V	49 GPa	strain	0.15 %
distance	0.15	$B \cdot n$	2.37 ‰
steel	30 kt	clear.	1.22 m

violated: neutron fluence.

0127 −107 %

coils	16	σ_{99}	1 037 MPa
conn./pair	7	max.	8 941 MPa
E_V	55 GPa	strain	0.49 %
distance	0.27	$B \cdot n$	3.86 ‰
steel	19 kt	clear.	1.79 m

violated: steel σ99, winding-pack strain, field error B·n.

0128 −556 %

coils	24	σ_{99}	611 MPa
conn./pair	11	max.	2 438 MPa
E_V	72 GPa	strain	0.32 %
distance	0.15	$B \cdot n$	1.99 ‰
steel	30 kt	clear.	1.07 m

violated: steel σ99, winding-pack strain, package space, nuclear heating, neutron fluence.

0129 −182 %

coils	24	σ_{99}	1 411 MPa
conn./pair	12	max.	6 464 MPa
E_V	69 GPa	strain	0.82 %
distance	0.24	$B \cdot n$	1.42 ‰
steel	26 kt	clear.	1.80 m

violated: steel σ99, winding-pack strain.

0130 −90 %

coils	16	σ_{99}	782 MPa
conn./pair	14	max.	2 286 MPa
E_V	52 GPa	strain	0.35 %
distance	0.23	$B \cdot n$	5.71 ‰
steel	27 kt	clear.	1.51 m

violated: steel σ99, winding-pack strain, field error B·n.

0131 −220 %

coils	20	σ_{99}	1 598 MPa
conn./pair	14	max.	20 042 MPa
E_V	151 GPa	strain	0.91 %
distance	0.20	$B \cdot n$	1.44 ‰
steel	25 kt	clear.	1.43 m

violated: steel σ99, winding-pack strain.

0132 −286 %

coils	16	σ_{99}	640 MPa
conn./pair	11	max.	3 689 MPa
E_V	64 GPa	strain	0.34 %
distance	0.25	$B \cdot n$	11.59 ‰
steel	21 kt	clear.	1.58 m

violated: steel σ99, winding-pack strain, field error B·n.

0133 −363 %

coils	20	σ_{99}	2 313 MPa
conn./pair	7	max.	362 657 MPa
E_V	24 GPa	strain	1.31 %
distance	0.26	$B \cdot n$	2.80 ‰
steel	23 kt	clear.	1.69 m

violated: steel σ99, winding-pack strain.

0134 −4 220 %

coils	24	σ_{99}	1 125 MPa
conn./pair	14	max.	2 886 MPa
E_V	106 GPa	strain	0.40 %
distance	0.13	$B \cdot n$	1.99 ‰
steel	32 kt	clear.	0.88 m

violated: steel σ99, winding-pack strain, package space, nuclear heating, neutron fluence.

0135 −272 %

coils	16	σ_{99}	1 862 MPa
conn./pair	9	max.	18 564 MPa
E_V	107 GPa	strain	0.94 %
distance	0.21	$B \cdot n$	3.40 ‰
steel	21 kt	clear.	1.82 m

violated: steel σ99, winding-pack strain, field error B·n.

0136* −15 %

coils	20	σ_{99}	577 MPa
conn./pair	13	max.	2 238 MPa
E_V	21 GPa	strain	0.30 %
distance	0.23	$B \cdot n$	1.62 ‰
steel	26 kt	clear.	1.84 m

violated: steel σ99.

0137 −22 %

coils	24	σ_{99}	612 MPa
conn./pair	6	max.	2 562 MPa
E_V	39 GPa	strain	0.29 %
distance	0.16	$B \cdot n$	0.94 ‰
steel	16 kt	clear.	1.57 m

violated: steel σ99.

0138 −56 %

coils	24	σ_{99}	780 MPa
conn./pair	12	max.	3 442 MPa
E_V	180 GPa	strain	0.45 %
distance	0.24	$B \cdot n$	0.66 ‰
steel	28 kt	clear.	1.81 m

violated: steel σ99, winding-pack strain.

0139 −506 %

coils	20	σ_{99}	2 862 MPa
conn./pair	11	max.	30 000 MPa
E_V	36 GPa	strain	1.82 %
distance	0.29	$B \cdot n$	2.82 ‰
steel	26 kt	clear.	2.02 m

violated: steel σ99, winding-pack strain.

0140 −31 %

coils	20	σ_{99}	656 MPa
conn./pair	13	max.	4 169 MPa
E_V	66 GPa	strain	0.34 %
distance	0.26	$B \cdot n$	1.46 ‰
steel	22 kt	clear.	1.76 m

violated: steel σ99, winding-pack strain.

0141 −124 %

coils	16	σ_{99}	884 MPa
conn./pair	13	max.	2 092 MPa
E_V	26 GPa	strain	0.32 %
distance	0.18	$B \cdot n$	6.73 ‰
steel	20 kt	clear.	1.41 m

violated: steel σ99, winding-pack strain, field error B·n.

0142 −145 %

coils	16	σ_{99}	830 MPa
conn./pair	14	max.	4 887 MPa
E_V	30 GPa	strain	0.38 %
distance	0.18	$B \cdot n$	7.34 ‰
steel	29 kt	clear.	1.52 m

violated: steel σ99, winding-pack strain, field error B·n.

0143 −52 %

coils	20	σ_{99}	761 MPa
conn./pair	10	max.	1 973 MPa
E_V	115 GPa	strain	0.28 %
distance	0.17	$B \cdot n$	1.25 ‰
steel	22 kt	clear.	1.34 m

violated: steel σ99.

0144 −297 %

coils	24	σ_{99}	1 778 MPa
conn./pair	11	max.	9 091 MPa
E_V	79 GPa	strain	1.19 %
distance	0.28	$B \cdot n$	4.80 ‰
steel	18 kt	clear.	2.13 m

violated: steel σ99, winding-pack strain, field error B·n.

0145* −79 %

coils	16	σ_{99}	426 MPa
conn./pair	14	max.	2 490 MPa
E_V	77 GPa	strain	0.20 %
distance	0.22	$B \cdot n$	5.37 ‰
steel	36 kt	clear.	1.37 m

violated: field error B·n.

0147 −406 %

coils	24	σ_{99}	465 MPa
conn./pair	8	max.	1 292 MPa
E_V	45 GPa	strain	0.17 %
distance	0.19	$B \cdot n$	2.02 ‰
steel	20 kt	clear.	1.10 m

violated: package space, neutron fluence.

0148 −128 %

coils	24	σ_{99}	1 141 MPa
conn./pair	13	max.	3 585 MPa
E_V	57 GPa	strain	0.53 %
distance	0.23	$B \cdot n$	0.74 ‰
steel	34 kt	clear.	1.61 m

violated: steel σ99, winding-pack strain.

0149* −67 %

coils	20	σ_{99}	636 MPa
conn./pair	14	max.	1 423 MPa
E_V	26 GPa	strain	0.25 %
distance	0.22	$B \cdot n$	5.02 ‰
steel	18 kt	clear.	1.56 m

violated: steel σ99, field error B·n.

0150 −370 %

coils	20	σ_{99}	2 129 MPa
conn./pair	7	max.	31 817 MPa
E_V	134 GPa	strain	1.41 %
distance	0.28	$B \cdot n$	6.81 ‰
steel	15 kt	clear.	2.07 m

violated: steel σ99, winding-pack strain, field error B·n.

0151 −120 %

coils	20	σ_{99}	1 014 MPa
conn./pair	8	max.	11 644 MPa
E_V	153 GPa	strain	0.66 %
distance	0.17	$B \cdot n$	2.03 ‰
steel	26 kt	clear.	1.47 m

violated: steel σ99, winding-pack strain.

0152 −126 %

coils	24	σ_{99}	1 024 MPa
conn./pair	14	max.	4 796 MPa
E_V	33 GPa	strain	0.68 %
distance	0.25	$B \cdot n$	0.99 ‰
steel	40 kt	clear.	1.66 m

violated: steel σ99, winding-pack strain.

0153 −241 %

coils	24	σ_{99}	1 706 MPa
conn./pair	11	max.	16 338 MPa
E_V	51 GPa	strain	1.01 %
distance	0.28	$B \cdot n$	2.63 ‰
steel	25 kt	clear.	1.94 m

violated: steel σ99, winding-pack strain.

0154 −413 %

coils	20	σ_{99}	2 565 MPa
conn./pair	10	max.	21 004 MPa
E_V	61 GPa	strain	1.33 %
distance	0.29	$B \cdot n$	5.82 ‰
steel	14 kt	clear.	2.22 m

violated: steel σ99, winding-pack strain, field error B·n.

0155 −233 %

coils	20	σ_{99}	1 607 MPa
conn./pair	6	max.	31 742 MPa
E_V	123 GPa	strain	1.00 %
distance	0.29	$B \cdot n$	2.28 ‰
steel	13 kt	clear.	2.20 m

violated: steel σ99, winding-pack strain.

0156 −108 %

coils	16	σ_{99}	623 MPa
conn./pair	11	max.	3 243 MPa
E_V	105 GPa	strain	0.31 %
distance	0.26	$B \cdot n$	6.25 ‰
steel	26 kt	clear.	1.66 m

violated: steel σ99, winding-pack strain, field error B·n.

0157 2 %

coils	20	σ_{99}	492 MPa
conn./pair	6	max.	1 658 MPa
E_V	158 GPa	strain	0.22 %
distance	0.22	$B \cdot n$	2.37 ‰
steel	18 kt	clear.	1.35 m

admissible.

0158 −444 %

coils	20	σ_{99}	2 486 MPa
conn./pair	6	max.	14 984 MPa
E_V	56 GPa	strain	1.63 %
distance	0.26	$B \cdot n$	2.72 ‰
steel	15 kt	clear.	2.11 m

violated: steel σ99, winding-pack strain.

0159 −284 %

coils	24	σ_{99}	1 922 MPa
conn./pair	9	max.	4 529 MPa
E_V	134 GPa	strain	0.73 %
distance	0.14	$B \cdot n$	2.62 ‰
steel	13 kt	clear.	1.29 m

violated: steel σ99, winding-pack strain.

0160 −88 %

coils	16	σ_{99}	941 MPa
conn./pair	6	max.	5 488 MPa
E_V	88 GPa	strain	0.51 %
distance	0.13	$B \cdot n$	3.31 ‰
steel	22 kt	clear.	1.57 m

violated: steel σ99, winding-pack strain, field error B·n.

0161 −205 %

coils	24	σ_{99}	1 102 MPa
conn./pair	14	max.	19 494 MPa
E_V	29 GPa	strain	0.91 %
distance	0.27	$B \cdot n$	1.66 ‰
steel	32 kt	clear.	1.94 m

violated: steel σ99, winding-pack strain.

0162 −451 %

coils	20	σ_{99}	2 755 MPa
conn./pair	13	max.	12 320 MPa
E_V	71 GPa	strain	1.15 %
distance	0.20	$B \cdot n$	2.52 ‰
steel	26 kt	clear.	1.26 m

violated: steel σ99, winding-pack strain, neutron fluence.

0163 −650 %

coils	24	σ_{99}	1 347 MPa
conn./pair	9	max.	6 590 MPa
E_V	45 GPa	strain	0.64 %
distance	0.16	$B \cdot n$	1.89 ‰
steel	29 kt	clear.	1.06 m

violated: steel σ99, winding-pack strain, package space, nuclear heating, neutron fluence.

0164 −262 %

coils	20	σ_{99}	1 810 MPa
conn./pair	7	max.	7 235 MPa
E_V	43 GPa	strain	0.84 %
distance	0.26	$B \cdot n$	2.14 ‰
steel	17 kt	clear.	1.99 m

violated: steel σ99, winding-pack strain.

0165 −765 %

coils	20	σ_{99}	659 MPa
conn./pair	8	max.	1 698 MPa
E_V	125 GPa	strain	0.25 %
distance	0.17	$B \cdot n$	4.01 ‰
steel	15 kt	clear.	1.05 m

violated: steel σ99, package space, nuclear heating, neutron fluence, field error B·n.

0166 −224 %

coils	20	σ_{99}	1 620 MPa
conn./pair	6	max.	7 156 MPa
E_V	32 GPa	strain	0.70 %
distance	0.20	$B \cdot n$	2.26 ‰
steel	15 kt	clear.	1.40 m

violated: steel σ99, winding-pack strain.

0167 −117 %

coils	24	σ_{99}	1 064 MPa
conn./pair	11	max.	4 862 MPa
E_V	30 GPa	strain	0.65 %
distance	0.23	$B \cdot n$	0.89 ‰
steel	34 kt	clear.	1.63 m

violated: steel σ99, winding-pack strain.

0168 −566 %

coils	16	σ_{99}	3 328 MPa
conn./pair	12	max.	87 021 MPa
E_V	31 GPa	strain	1.63 %
distance	0.21	$B \cdot n$	4.40 ‰
steel	19 kt	clear.	1.79 m

violated: steel σ99, winding-pack strain, field error B·n.

0169* 29 %

coils	24	σ_{99}	308 MPa
conn./pair	13	max.	717 MPa
E_V	183 GPa	strain	0.15 %
distance	0.20	$B \cdot n$	2.13 ‰
steel	28 kt	clear.	1.39 m

admissible.

0170 −153 %

coils	16	σ_{99}	1 264 MPa
conn./pair	9	max.	4 001 MPa
E_V	21 GPa	strain	0.51 %
distance	0.12	$B \cdot n$	5.58 ‰
steel	17 kt	clear.	1.53 m

violated: steel σ99, winding-pack strain, field error B·n.

0171 −66 %

coils	24	σ_{99}	829 MPa
conn./pair	10	max.	1 952 MPa
E_V	102 GPa	strain	0.33 %
distance	0.20	$B \cdot n$	0.56 ‰
steel	17 kt	clear.	1.54 m

violated: steel σ99, winding-pack strain.

0172 −204 %

coils	16	σ_{99}	1 522 MPa
conn./pair	6	max.	4 140 MPa
E_V	67 GPa	strain	0.70 %
distance	0.22	$B \cdot n$	3.41 ‰
steel	16 kt	clear.	1.64 m

violated: steel σ99, winding-pack strain, field error B·n.

0174 −228 %

coils	16	σ_{99}	938 MPa
conn./pair	10	max.	9 438 MPa
E_V	20 GPa	strain	0.41 %
distance	0.21	$B \cdot n$	9.84 ‰
steel	18 kt	clear.	1.55 m

violated: steel σ99, winding-pack strain, field error B·n.

0175 −114 %

coils	16	σ_{99}	976 MPa
conn./pair	10	max.	8 082 MPa
E_V	53 GPa	strain	0.56 %
distance	0.27	$B \cdot n$	6.41 ‰
steel	17 kt	clear.	2.15 m

violated: steel σ99, winding-pack strain, field error B·n.

0176* −91 %

coils	20	σ_{99}	434 MPa
conn./pair	14	max.	2 674 MPa
E_V	165 GPa	strain	0.22 %
distance	0.19	$B \cdot n$	4.86 ‰
steel	32 kt	clear.	1.20 m

violated: neutron fluence, field error B·n.

0177 −73 %

coils	24	σ_{99}	867 MPa
conn./pair	10	max.	4 068 MPa
E_V	129 GPa	strain	0.45 %
distance	0.24	$B \cdot n$	1.38 ‰
steel	19 kt	clear.	1.78 m

violated: steel σ99, winding-pack strain.

0178 −65 %

coils	20	σ_{99}	827 MPa
conn./pair	8	max.	2 080 MPa
E_V	36 GPa	strain	0.40 %
distance	0.14	$B \cdot n$	2.44 ‰
steel	22 kt	clear.	1.47 m

violated: steel σ99, winding-pack strain.

0179 −222 %

coils	16	σ_{99}	1 612 MPa
conn./pair	7	max.	12 248 MPa
E_V	28 GPa	strain	0.71 %
distance	0.28	$B \cdot n$	4.34 ‰
steel	16 kt	clear.	1.88 m

violated: steel σ99, winding-pack strain, field error B·n.

0180 −278 %

coils	16	σ_{99}	1 890 MPa
conn./pair	13	max.	10 210 MPa
E_V	35 GPa	strain	0.94 %
distance	0.27	$B \cdot n$	1.74 ‰
steel	22 kt	clear.	1.99 m

violated: steel σ99, winding-pack strain.

0181 −56 %

coils	24	σ_{99}	782 MPa
conn./pair	9	max.	2 109 MPa
E_V	189 GPa	strain	0.32 %
distance	0.15	$B \cdot n$	2.14 ‰
steel	18 kt	clear.	1.23 m

violated: steel σ99, winding-pack strain, neutron fluence.

0182 −383 %

coils	16	σ_{99}	2 258 MPa
conn./pair	14	max.	89 455 MPa
E_V	49 GPa	strain	1.45 %
distance	0.24	$B \cdot n$	10.61 ‰
steel	18 kt	clear.	1.48 m

violated: steel σ99, winding-pack strain, field error B·n.

0183 −211 %

coils	20	σ_{99}	1 555 MPa
conn./pair	6	max.	11 544 MPa
E_V	83 GPa	strain	0.85 %
distance	0.20	$B \cdot n$	2.36 ‰
steel	15 kt	clear.	1.57 m

violated: steel σ99, winding-pack strain.

0184 −124 %

coils	24	σ_{99}	361 MPa
conn./pair	9	max.	799 MPa
E_V	122 GPa	strain	0.16 %
distance	0.13	$B \cdot n$	1.87 ‰
steel	23 kt	clear.	1.18 m

violated: neutron fluence.

0186 −774 %

coils	20	σ_{99}	512 MPa
conn./pair	8	max.	1 244 MPa
E_V	34 GPa	strain	0.20 %
distance	0.14	$B \cdot n$	2.76 ‰
steel	24 kt	clear.	1.04 m

violated: steel σ99, package space, nuclear heating, neutron fluence.

0187 −119 %

coils	16	σ_{99}	1 094 MPa
conn./pair	13	max.	4 736 MPa
E_V	58 GPa	strain	0.51 %
distance	0.16	$B \cdot n$	1.73 ‰
steel	34 kt	clear.	1.77 m

violated: steel σ99, winding-pack strain.

0188 −151 %

coils	20	σ_{99}	492 MPa
conn./pair	12	max.	884 MPa
E_V	45 GPa	strain	0.20 %
distance	0.15	$B \cdot n$	1.74 ‰
steel	25 kt	clear.	1.17 m

violated: neutron fluence.

0189 −304 %

coils	16	σ_{99}	1 510 MPa
conn./pair	7	max.	17 307 MPa
E_V	99 GPa	strain	1.21 %
distance	0.30	$B \cdot n$	9.07 ‰
steel	20 kt	clear.	2.08 m

violated: steel σ99, winding-pack strain, field error B·n.

0190 −71 %

coils	24	σ_{99}	857 MPa
conn./pair	14	max.	4 610 MPa
E_V	22 GPa	strain	0.44 %
distance	0.18	$B \cdot n$	2.44 ‰
steel	31 kt	clear.	1.22 m

violated: steel σ99, winding-pack strain, neutron fluence.

0192 −367 %

coils	16	σ_{99}	1 836 MPa
conn./pair	9	max.	62 386 MPa
E_V	40 GPa	strain	1.40 %
distance	0.25	$B \cdot n$	12.61 ‰
steel	15 kt	clear.	1.86 m

violated: steel σ99, winding-pack strain, field error B·n.

0193 −186 %

coils	16	σ_{99}	585 MPa
conn./pair	11	max.	1 284 MPa
E_V	76 GPa	strain	0.25 %
distance	0.13	$B \cdot n$	8.57 ‰
steel	16 kt	clear.	1.45 m

violated: steel σ99, field error B·n.

0194 −119 %

coils	20	σ_{99}	1 097 MPa
conn./pair	12	max.	2 077 MPa
E_V	24 GPa	strain	0.39 %
distance	0.21	$B \cdot n$	1.63 ‰
steel	19 kt	clear.	1.49 m

violated: steel σ99, winding-pack strain.

0195 −529 %

coils	24	σ_{99}	359 MPa
conn./pair	10	max.	932 MPa
E_V	80 GPa	strain	0.15 %
distance	0.12	$B \cdot n$	2.36 ‰
steel	22 kt	clear.	1.08 m

violated: package space, nuclear heating, neutron fluence.

0196 −111 %

coils	16	σ_{99}	672 MPa
conn./pair	8	max.	1 529 MPa
E_V	200 GPa	strain	0.29 %
distance	0.15	$B \cdot n$	6.33 ‰
steel	15 kt	clear.	1.63 m

violated: steel σ99, field error B·n.

0198 −286 %

coils	20	σ_{99}	1 930 MPa
conn./pair	9	max.	18 667 MPa
E_V	35 GPa	strain	0.93 %
distance	0.18	$B \cdot n$	2.45 ‰
steel	18 kt	clear.	1.37 m

violated: steel σ99, winding-pack strain.

0199 −71 %

coils	24	σ_{99}	855 MPa
conn./pair	7	max.	2 060 MPa
E_V	22 GPa	strain	0.37 %
distance	0.23	$B \cdot n$	1.48 ‰
steel	16 kt	clear.	1.50 m

violated: steel σ99, winding-pack strain.

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